Research Report

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Sustaining Agile Operations

Volume 2, Energy and Water Technology Survey

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Published by the RAND Corporation, Santa Monica, Calif.

RR-A3582-2

About This Report

This report provides an overview of advancements in energy and water technologies that might have the potential to support U.S. Air Force operations in contested environment. This research was part of a larger study of how energy and water availability, management, and supply chain risk will affect the Air Force's ability to conduct rapid and agile operations in combat. This is a follow-on effort to Kristin Van Abel and colleagues, *Sustaining Agile Operations: Volume 1, An Approach to Evaluating Energy and Water Resilience Options and Setting Investment Priorities Across the Theater*, Not available to the general public.

The research reported here was sponsored by the Deputy Assistant Secretary of the Air Force Infrastructure, Energy, and Environment and prepared under contract FA7014-22-D-0001 as part of a PAF Resource Management fiscal year 2025 project, "Sustaining and Fueling the Great Power Competition (GPC) Fight."

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Acknowledgments

We appreciate the support we received from many organizations within the U.S. Air Force. Specifically, we thank Ms. Nancy Balkus, SAF/IEE, for sponsoring this work and Mr. Doug Tucker and Ms. Judith Willis, SAF/IEE, for their continued help and guidance throughout the analysis.

Additionally, we thank the many personnel from across Air Force Materiel Command who met with us and provided valuable insights including the Air Force Civil Engineering Center (AFCEC) Office of Energy Assurance, AFCEC Expeditionary Engineering, the Air Force

Research Lab (AFRL) Energy Office and AFWERX. We thank those personnel across the Indo-Pacific theater who provided perspectives that informed the analysis including Headquarters Pacific Air Forces, 18 Fighter Wing, and 51 Fighter Wing.

Finally, at the RAND Corporation, we thank Mr. James Silva for his input and counsel, Beth Grill and Maj Gen (r) Stephen Oliver, Jr., for their reviews, and Barbara Bicksler for her efforts to make this report more logical, readable, and digestible. We also thank Stephanie Young and Anna Jean Wirth for their oversight of the project.

Help and insights received from those acknowledged above should not be taken to imply that they concur with the views expressed in this document. The analysis and conclusions in the document are solely the responsibility of the authors.

Summary

Issue

The U.S. Air Force (USAF) endeavors to enhance its capabilities to execute rapid and flexible operations in contested environment. Matters associated with logistics and sustainment play a significant role in the success of such operations, including the availability of energy and water. Technological advancements can play a role in helping the USAF more efficiently and effectively meet its energy and water requirements during combat.

Approach

This work draws heavily on prior RAND research conducted for Pacific Air Forces/A4 in 2021 that investigated a large selection of emerging technologies for agile combat employment (ACE) concepts of operations, some of which were energy and water related. In this prior work, the authors conducted a technology evaluation that assessed technical maturity, suitability for agile combat operations, degree of logistics burden, among other factors.[1] This report presents updates to a select set of electricity generation, energy storage, and water technology evaluations included in the original work from 2021. A subset of the updated technologies is applied to energy and water demands examined in Volume 1 of this two-part report series.

Observations

- Advancements in existing power technologies are making near-term deployment more feasible.
  - Batteries, vehicle-to-grid systems, and airborne wind turbine technology have seen notable progress in cost and performance, positioning them closer to operational readiness.
  - Airborne wind turbine technology has transitioned from concept to credible prototypes, showing promise for expeditionary applications.
- Transformational technologies are progressing but remain at an early stage.
  - Fusion research is expanding globally, with academic and industry investment aimed at increasing the practicality of reactors, though commercial viability remains distant.

[1] For details see: Dahlia Anne Goldfeld, Richard J. Mason, Samantha McBirney, Kristin F. Lynch, Edward Geist, Jared Mondschein, Monica Rico, *Technologies to Enhance Logistics Support of a High-End Fight and Agile Combat Employment*: Vol. 2, Technology Survey, RAND Corporation, RR-A978-2, 2023. Not available to the general public.

  - Nuclear microreactors and small modular reactors are under development and testing, but still face challenges in regulation, sustainment, and personnel requirements.
  - Novel water treatment technologies, such as capacitive deionization[2], are emerging in the commercial sector but need further maturity before reliable military use.
  - Vehicle-to-grid systems have become more widely used commercially, which have led to strategies for adoption and deployment.

- Established water and energy production alternatives are improving and could enhance operational efficiency in the near term.
  - Reverse osmosis and water desalination technologies which are already in use have improved potable water production could be advantageous for remote missions.
  - Atmospheric water harvesting systems have matured and have high potential to provide potable water in very austere environments, even when electricity is not available.
  - Airborne wind turbines have promising portability and a relatively low levelized cost of energy, which makes it a good candidate as an alternative for traditional wind power in an operational environment, modeled in Chapter 3 in Volume 1 of this report.
  - Photovoltaics panels, an existing asset within DOW, have improved in efficiency and chemistry, making solar farms more cost effective. The compared operational efficiency of alternative solar panels assessed were also modeled in Volume 1.

These observations are solely based on findings from updating the technology survey and are independent of the recommendations given in Volume 1 of this report series.

[2] Capacitive deionization (CDI) is an electrochemical technique that removes salt and charged contaminants from water by applying a voltage across porous electrodes, causing ions to absorb into electrical double layers.

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Chapter 1. Introduction

As part of Project AIR FORCE's assessment of how energy and water availability, management, and supply chain risk will affect the U.S. Air Force's (USAF's) ability to execute rapid and flexible operations in a contested environment, we examined select technologies relevant to these topics. This work draws heavily on prior RAND research conducted for Pacific Air Forces (PACAF)/A4 in 2021 that investigated a large selection of emerging technologies for agile combat employment (ACE) concepts of operations, some of which were energy and water related. In this prior work, the authors conducted a technology evaluation that assessed technical maturity, suitability for ACE, degree of logistics burden, among other factors.[3] Here, we present updates to a select set of electricity generation, energy storage, and water technology evaluations included in the original work. We only reassessed energy and water-related technologies that appeared operationally useful and technologically promising in the original work. Volume 2 of this series is designed to inform stakeholders of alternatives for water and energy technologies that have matured enough to potentially support ACE operations for sortie generation as detailed in Volume 1.

The technologies along with the primary metrics that we used to evaluate them are listed in Table 1.1. We define "deployability" as a "meta-metric" to rate a technology on how easily it could be used in an ACE operation. Deployability is itself comprised of set of metrics including weight, volume, ease of set up, and packaging and palletization. Although technical maturity is a very important metric for all technologies reviewed, it is particularly important for the technologies that were the least mature in the previous work (e.g., airborne wind turbines or nuclear fusion). We did not expect, and did not find, that more mature technologies, such as photovoltaics (PV), would experience giant gains in maturity because these systems were already commercially available and widely used in 2021.

Our results are summarized in Figure 1.1 and elaborated on later chapters of this report. The columns represent different levels of technical maturity and are a loosely consolidated version of the Technology Readiness Level scale.[4] The rows in the figure denote two different levels of technological evolution. None of the technologies that we re-evaluated were unchanged (this

[3] For details see: Dahlia Anne Goldfeld, Richard J. Mason, Samantha McBirney, Kristin F. Lynch, Edward Geist, Jared Mondschein, Monica Rico, *Technologies to Enhance Logistics Support of a High-End Fight and Agile Combat Employment*: Vol. 2, Technology Survey, RAND Corporation, RR-A978-2, 2023, Not available to the general public.

[4] See, for instance, Catherine G. Manning, "Technology Readiness Levels." NASA, September 27, 2023, UNCLASSIFIED. As of March 1, 2025: https://www.nasa.gov/directorates/somd/space-communications-navigation-program/technology-readiness-levels/

would have occurred only if a technology had been abandoned by the research community or if commercialization had failed).

Table 1.1. Important Metrics Used to Evaluate Technologies

Technology	Metrics
Airborne wind turbines	• Technical maturity • Energy conversion • Portability • Deployability
Photovoltaics	• Conversion rate • Durability • Deployability
Fuel cells	• Technical maturity • Access to hydrogen
Nuclear fusion	• Technical maturity • Cost
Portable nuclear reactors	• Technical maturity • Deployability • Safety
Batteries	• Energy density (how much energy can be stored per unit weight)
Ultracapacitors	• Energy density • Charging and recharging cycle speed
Hydrogen energy storage	• Technical maturity • Density • Safety
Vehicle-to-grid systems	• Technical maturity
Reverse osmosis (RO) (expeditionary systems)	• Energy efficiency • Deployability and ruggedness
Water filtration and decontamination	• New methods and materials • Effectiveness in removing chemicals, radioactive materials, and neutralizing bacteria viruses and fungi
Capacitive deionization	• Technical maturity • Energy efficiency • Ability to work for saltier than brackish water
Atmospheric water harvesting	• Deployability • Water extraction efficiency at different humidity levels

Figure 1.1. Summary of the Level of Technological Advancement Over the Last Five Years

	Deployment Ready	Operational Environment Prototype	Commercial Demonstration	Lab - Validated Prototype	Experimental Proof of Novel Concept
Incremental Changes	Photovoltaics Reverse Osmosis Water Filtration and Decontamination Ultracapacitors	Fuel Cells Hydrogen Energy Storage	Portable Nuclear Reactors	Nuclear Fusion Capacitive Deionization	
Major Advances	Batteries Vehicle-to-Grid Systems Airborne Wind Turbines	Atmospheric Water Harvesting			

Energy
Water

All the technologies we evaluated experienced improvements along at least some of the relevant metrics. We do not have a rigorous definition of incremental changes versus major advances. We made a judgment call based on our own understanding of each technology and what fraction of capability recent improvements seem to represent. The more mature a technology is, the less likely it is to experience a major advance in just a few years. As a popular example, consider the iPhone. The first iPhone represented a huge leap over the PalmPilot (which was the state-of-the-art smart phone). The next few generations of iPhones each had large improvements in form and function, as evidenced by the fact that very large numbers of users would buy a new iPhone each year. However, over time, consumers became more tolerant of skipping an iPhone generation, and today people will keep their iPhones for years before replacing them. This change in behavior occurred not because the newest generation phone does not have new features, but because they simply are not compelling enough to many people to warrant an outlay of $1,500. For the iPhone, one could use this measure of consumer demand as a measure of what fraction of total iPhone capacity occurred between the prior generation and the most current.

We do not have this kind of mass consumer data for the technologies evaluated in this volume. However, a technology like airborne wind turbines has overcome a very large hurdle in the last five years and is now commercialized. Atmospheric water harvesters were already commercially available in 2021, but the market has increased substantially since then. PV continues to slowly come down in price, and there are new chemistries on the market, but fundamentally the solar energy conversion rates have not changed very much. For this reason, we consider PV to have experienced incremental changes. Figure 1.1 illustrates our determination of where each technology lies on the technical maturity-technological evolutionaxis. The remainder of this volume contains details about each technology reassessed from RAND’s 2021 study, specifically technologies relevant to meeting the power and water requirements of PACAF/A4 and discusses considerations for the utility of these technologies to the USAF in the ACE environment.

Chapter 2. Electricity Generation and Storage Technologies

We considered nine existing or emerging technologies that generate or store energy and could be used to support ACE operations. We provide a general description of technology, document how the technology evolved since the 2021 survey, explain key considerations for USAF employment of the technology, and provide information about a rough order of magnitude (ROM) cost for each of the technologies in the sections below.

Airborne Wind Turbines

Airborne wind power uses tethered kites to harness high-altitude winds for electricity generation, offering portable and deployable energy solutions for remote and scarce environments. Recent advancements have improved efficiency, scalability, and commercial viability, making technology increasingly practical for applications such as agile combat operations, disaster relief, and off-grid energy systems.

SkySails Power introduced the first commercially available airborne wind energy system in 2021, delivering 80 to 200 kW of power in a compact 30-foot container, with a megawatt-class version planned for 2024.[5] Kitepower offers a 100-kW system housed in a 20-foot container. These systems operate autonomously in most weather conditions, achieving rated power capacities approximately 75 percent of the time, though they must be grounded during thunderstorms or calm winds. Regulatory requirements mandate an 800-meter radius of unobstructed space for each kite, with systems spaced 400 meters apart.[6]

Recent innovations have expanded the technology's capabilities. Jiangxi's airborne wind power generator, a helium-filled S500 blimp known as the Strastospheri Floating Wind Power System developed by Beijing Linyi Yuchuan Energy Technology Co. broke a world record for generating 50 kW of power 500 meters above ground, demonstrating the potential to generate windpower at greater altitudes. Windlift is developing systems tailored for military and disaster relief applications, emphasizing portability and rapid deployment. Advances in automation have improved operational reliability, and the cost of energy is now competitive with fixed wind turbines of similar capacity.[7]

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[5] SkySails Power, "Onshore Unit PN-14," webpage, 2024. As of August 24, 2025: https://skysails-power.com/onshore-unit-pn-14/

[6] Jochem Weber, Melinda Marquis, Aubryn Cooperman, Caroline Draxl, Rob Hammond, Jason Jonkman, Alexsandra Lemke, Anthony Lopez, Rafael Mudafort, Mike Optis, Owen Roberts, and Matt Shields, National Renewable Energy Laboratory, *Airborne Wind Energy*, National Renewable Energy Laboratory, NREL/TP-5000-79992, 2021. As of September 15, 2025: https://docs.nrel.gov/docs/fy21osti/79992.pdf

[7] Niu Yuhan, "Jiangxi Airborne Wind Power Generator Breaks World Record," Dialogue Earth, January 16, 2025; IOP Science, "Advances in Airborne Wind Power Technology," Environmental Research Letters, 2024.

Considerations for the USAF for the Employment of Airborne Wind Turbines

For the USAF, airborne wind power offers notable advantages in austere environments, reducing reliance on fuel resupply missions and enhancing mission sustainability. Systems like Skysails are ideal for forward operating bases, while automated kite descent mechanisms mitigate airspace conflicts.

However, airspace integration presents a critical operational challenge. Each airborne kite operates within an 800-meter radius and at altitudes of several hundred meters, which can interfere with flight operations if positioned near active runways or flight paths. Current airborne systems rely on automatic descent mechanisms, where the kite is reeled in and landed within minutes—typically three to five minutes under normal conditions—if flight operations need to clear the airspace or adverse weather approaches. However, these systems are not fail-proof: mechanical failures in the winch or kite control could result in entanglement or uncontrolled descent, posing safety hazards. Because of these risks, deployment locations would likely need to be at least 1 to 2 kilometers away from runways and approach or departure corridors, with defined deconfliction procedures like temporary unmanned aerial systems operations.[8]

Training and support are also required. USAF personnel are not currently trained to deploy, maintain, or troubleshoot airborne wind energy systems, and these skill sets differ from conventional generator maintenance. While automation reduces routine human input, initial setup, fault recovery, and safe airspace integration require a dedicated support team.

Cost considerations are mixed: Operational energy costs (per kilowatt-hour [kWh]) are now competitive with similar-sized fixed wind turbines and lower than sustained fuel supply in remote theaters, but upfront capital costs and the training burden represent a non-trivial investment. It was estimated that SkySails' Venyo, their most current commercial model, costs between $1.7 million to $2.8 million as an overall investment, with an estimated levelized cost of energy around $0.06/kWh.

In short, airborne wind power offers real benefits for ACE—logistics risk reduction, local energy generation, and containerized mobility—but also introduces unique challenges in airspace management, safety, personnel readiness, and upfront cost. Continued investment in research and development is essential to further enhance the technology's reliability and adoption.

Photovoltaics

PV devices remain a cornerstone of renewable energy technology, converting sunlight into electrical energy through materials such as silicon and gallium arsenide. However, recent advancements in PV technology have introduced new materials, designs, and applications that

[8] Michael Akers, "Airspace Deconfliction," *Risk Management Magazine*, June 6, 2021. As of September 15, 2025: https://safety.army.mil/MEDIA/Risk-Management-Magazine/ArtMID/7428/ArticleID/6958/Airspace-Deconfliction?

could enhance the operational capabilities of the USAF. These advancements include improvements in efficiency, reductions in size and weight, cost reductions, enhanced durability, and the integration of energy storage systems. Given the state of the art in PV, it is possible that this technology could exist in support of the energy demand in combat operations.

The efficiency of PV systems has significantly improved with the introduction of new materials. Perovskite photovoltaics illustrate this trend, positioning themselves as a serious competitor to silicon-based cells—achieving efficiencies ranging from 18 to 30 percent compared with the 16 percent efficiency of standard silicon cells.[9] For example, First Solar, a U.S. solar cell manufacturer recently released their Series 7 TR1, which are made from thin film cadmium telluride and are quoted to be 19.7 percent efficient.[10] Another example is Hanwha's Qcells, a Korean-based manufacturer who chose to advance the usage of silicon, achieved an efficiency of 28.6 percent, while maintaining a small area of 0.033 square meters.[11] Multijunction solar cells, which stack multiple light-absorbing layers tuned to different portions of the solar spectrum, have achieved efficiencies approaching 50 percent—well beyond the theoretical limit of single-junction devices.[12] These cells are particularly suited for high-performance applications, such as powering unmanned aerial vehicles (UAVs) or satellites. Organic photovoltaics have advanced significantly, with state-of-the-art research cells now achieving around 19 to 20 percent certified efficiency, while the best large-area modules have reached about 14.5 percent.[13] Organic photovoltaics are highly flexible and durable, making them suitable for integration into wearable devices or lightweight structures.

In addition to improvements in efficiency, PV systems have also become more compact and lightweight. Panels that are both foldable and rollable now weigh as little as 0.38 pounds for a 5 W output and 11.1 pounds for a 186 W output. However, newer designs have further reduced weight while increasing power density, enabling easier transport and rapid deployment in austere environments.[14] A newer design of PV panel by Merlin Solar produces a 240 W monocrystalline cell available at around $900 per unit.[15] Cost reductions have also been significant, with the

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[9] National Renewable Energy Laboratory, "Perovskite Solar Cells," webpage, undated. As of August 24, 2025: https://www.nrel.gov/pv/perovskite-solar-cells

[10] First Solar, "Series 7: A High-Quality Thin Film CdTe Module Made in America, for America," webpage, undated. As of September 15, 2025: https://www.firstsolar.com/Products/Series-7?

[11] Hanwha, "Hanwha Solutions Qcells Division," webpage, undated. As of September 8, 2025: https://www.hanwha.com/companies/hanwha-solutions-qcells-division.do

[12] National Renewable Energy Laboratory, *III-V Multijunction Solar Cells: High-Efficiency Photovoltaics for Space and Terrestrial Applications*, National Renewable Energy Laboratory, NREL/TP-85315, 2023.

[13] National Renewable Energy Laboratory, "Best Research-Cell Efficiency Chart," webpage, July 15, 2025. As of August 24, 2025: https://www.nrel.gov/pv/cell-efficiency

[14] National Renewable Energy Laboratory, "Photovoltaic Performance," webpage, August 3, 2025. As of September 8, 2025: https://www.nrel.gov/pv/real-time-photovoltaic-solar-resource-testing

[15] Merlin Solar, homepage, undated. As of September 8, 2025: https://www.merlinsolar.com/

median cost per watt for large-scale solar installations dropping to below $1/W compared to $1.2/W.[16] This reduction is driven by economies of scale and advancements in manufacturing. Emerging technologies, such as perovskite and organic photovoltaics cells, promise further reductions in manufacturing costs because of simpler production processes.[17]

Equally important are strides in grid integration, where improved storage technologies are beginning to mitigate the intermittency of solar power. By stabilizing output during periods of low sunlight, these solutions strengthen reliability and reduce dependence on conventional backup sources.[18] Durability has also improved, with advances in encapsulation techniques making flexible solar panels more resistant to environmental stressors such as extreme temperatures, humidity, and mechanical wear.[19] These enhancements ensure longer operational lifespans in challenging environments.

Considerations for the USAF for the Employment of Photovoltaic Systems

Advancements in PV technology could have important implications for USAF operations. Deployable PV systems with integrated storage can reduce reliance on diesel generators, which are costly to transport and vulnerable to supply chain disruptions, particularly advantage for forward operating bases and remote installations. Lightweight, flexible PV panels can now be integrated into shelters, vehicles, and even uniforms, providing portable power sources for personnel and equipment. This enhances operational flexibility and reduces logistical burdens.

High efficiency, multijunction cells can extend the operational range and endurance of UAVs, enabling longer reconnaissance missions. Similarly, these cells can improve the power generation capabilities of satellites, supporting advanced communication and surveillance systems. The continued decline in PV costs, coupled with reduced fuel consumption, offers significant economic benefits. For example, replacing diesel generators with PV systems at a battalion-sized outpost could save tens of thousands of gallons of fuel annually, as previously noted. Improved durability and energy storage integration make PV systems more reliable in extreme conditions, ensuring uninterrupted power supply during critical missions.

Although advancements in PV technology offer high efficiency, lighter weight, and reduced costs, these systems still require available space and setup time, which can be a challenge in contested or space-constrained environments. Deployable PV systems are ideal for mobile or short-duration operations but add daily tasks for deployment, alignment, and protection.

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[16] Katherine Antonio, "Today in Energy: New Solar Plants Expected to Support Most U.S. Electric Generation Growth," U.S. Energy Information Administration, January 24, 2025.

[17] University of Maryland, Center for Global Sustainability, "From Potential to Reality: Enhancing Solar PV Efficiency in China," news release, August 21, 2024.

[18] Financial Times, "Can the Solar Industry Keep the Lights On?" July 22, 2024. https://www.ft.com/content/9c53f696-6e32-4349-bd18-f4d431e2a577

[19] QCells USA, "Solar Panels: Q.ANTUM Technology That Makes a Difference," webpage, undated. https://us.qcells.com/complete-energy-solutions/solar-panel/

Permanent installations, on the other hand, eliminate repeated setup and offer long-term energy security, but require host-nation approval, clearly defined ownership, and ongoing maintenance agreements. Without established maintenance or security provisions, permanent systems risk falling into disrepair or becoming politically sensitive.

A hybrid approach may be optimal: deployable PV-plus-storage systems for mobile forces and permanent or semi-permanent arrays for enduring locations, potentially under dual-use agreements allowing host nations to benefit from the infrastructure when USAF forces are absent. This arrangement maximizes operational flexibility and reduces reliance on fuel convoys but also demands careful planning around logistics, security, and governance to avoid capability gaps.

Fuel Cells

Advancements in fuel cell technology during the past five years have been driven by national decarbonization goals, cost reduction pressures, and integration with renewable energy systems. Rather than relying on combustion, fuel cell technology draws on electrochemical reactions to transform the chemical energy stored in fuels like hydrogen or natural gas directly into electrical power. Prior RAND work considered the use of fuel cell technology as a major advantage for non-conventional power generation in PACAF operations.[20] At that time, state-of-the-art fuel cells (mainly hydrogen-based) were on the verge of becoming more efficient, clean, scalable, and reliable—claims that are still viable. Today, research and commercial efforts have expanded beyond hydrogen–based fuel cells toward solid oxide fuel cells (SOFCs) and direct ammonia fuel cell (DAFC) technologies, which offer unique advantages to fuel cells' operating constraints.

Commercial efforts are driving significant progress in hydrogen fuel cell technology. For example, Honda, in collaboration with General Motors, has developed next-generation fuel cells that double durability while reducing costs by two-thirds when compared with earlier models.[21] Weighing 250 kilograms, the next-generation fuel cell by Honda produces a net output of 150 kW, much higher than a fuel cell of similar weight produced by Toyota in 2021.[22] These systems are designed for heavy-duty vehicles, such as trucks and buses, as well as stationary power

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[20] Dahlia Anne Goldfeld, Kristin F. Lynch, John G. Drew, Richard Mason, Samantha McBirney, and Jared Mondschein, *Technologies to Enhance Logistics Support of a High-End Fight and Agile Combat Employment: Vol. 2*, RAND Corporation, RR-A978-1, 2023. Not available to the general public.

[21] Honda Motor Co., Ltd., "Honda and GM to Jointly Produce Next-Generation Fuel Cell System," webpage, February 19, 2025. As of September 15, 2025: https://global.honda/en/newsroom/news/2025/c250219eng.html?

[22] Lawrence Ulrich, "Honda's Next-Gen Hydrogen Fuel Cell Is Cheaper, More Durable," *IEEE Spectrum*, April 22, 2025. As of September 16, 2025: https://spectrum.ieee.org/honda-fuel-cell

applications, with scalability to support broader adoption across industries.[23] Hydrogen fuel cells are increasingly being deployed to power data centers, such as Edge Cloud Link, a data center solution company in Mountain View, California, that provides infrastructure for high-performance computing.[24]

SOFC's development has pushed the boundaries of fuel-cell technology, increasing the operation temperature to 600 to 1,000°C.[25] This allows for internal fuel reforming, which reduces the need for external processing, thus simplifying the circuit design.[26] SOFCs reach efficiencies of up to 60 percent as standalone systems and exceed 85 percent in combined heat and power setups, performance that makes them appealing for residential, industrial, and grid-scale applications.[27] Additionally, SOFCs are known for their durability, with lifespans that exceed those of many other fuel cell types, and, because of their modular scalability, they can be tailored to diverse energy needs, from small-scale installations to large industrial systems. One illustration of SOFC scalability is OxEon Energy, which received a $3.6 million grant in 2024 to expand its solid oxide electrolysis manufacturing capacity.[28] OxEon's solid oxide electrolysis technology first demonstrated its capabilities on NASA's Mars Perseverance Rover as part of the Mars Oxygen In Situ Resource Utilization Experiment (MOXIE), and the company has since adapted and scaled the device for use on Earth. This large-scale SOFC facility intends to produce around 25 MW annually.

DAFCs, on the other hand, use ammonia directly as a fuel, bypassing the need for hydrogen production and storage infrastructure. Ammonia is an energy-dense, easily transportable fuel that can be synthesized using renewable energy sources, making it a low-carbon alternative. DAFCs offer efficiencies comparable to hydrogen fuel cells while reducing the carbon footprint associated with fuel production. GenCell Energy's GenCell Fox (2022) illustrates this approach: the system pairs a patented ammonia-cracking unit that produces hydrogen on demand with the company's hydrogen fuel cell generators, and GenCell claims that just two 12-ton ammonia

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[23] Honda Motor Co., Ltd., "Honda Presents World Premiere Production Model of "CR-V e:FCEV" at H2 & FC EXPO Tokyo," news release, February 28, 2024. As of September 16, 2025: https://global.honda/en/newsroom/news/2024/4240228eng.html

[24] Rich Miller, "ECL Advances its Vision for Hydrogen-Powered Data Centers," Data Center Frontier, June 25, 2024. As of September 15, 2025: https://www.datacenterfrontier.com/energy/article/55090918/ecl-advances-its-vision-for-hydrogen-powered-data-centers

[25] Jingjing Li, Junhan Cheng, Yubing Zhang, Zhonghao Chen, Mahmoud Nasr, Mohamed Farghali, David W. Rooney, Pow-Seng Yap, and Ahmed I. Osman, "Advancements in Solid Oxide Fuel Cell Technology: Bridging Performance Gaps for Enhanced Environmental Sustainability," Advanced Energy and Sustainability Research, October 12, 2024. As of September 15, 2025: https://doi.org/10.1002/aesr.202400132

[26] ACS Publications, "Advances in Solid Oxide Fuel Cells: Materials, Designs, and Applications," *Chemical Reviews*, 2024.

[27] Sanjeev Mukerjee, "Using Solid-State Batteries and Hydrogen Fuel Cells to Power a Cleaner Future," Northeastern University, Center for Research Innovation, 2023.

[28] Oxeon Energy, "Oxeon Energy Wins $36 Million Hydrogen Funding from U.S. Department of Energy," news release, March 14, 2024.

tanks can supply enough fuel for continuous operation at rated power around the clock for an entire year.[29]

Considerations for the USAF for the Employment of Fuel Cell Technologies

Fuel cell technology could significantly enhance ACE operations by providing reliable, scalable, and clean power solutions for distributed and austere environments. Their modular design supports rapid deployment and adaptability to varying power demands, while their quiet operation and low emissions improve operational security and environmental compliance. Additionally, fuel cells' durability and ability to integrate with renewable energy systems can increase the resilience of infrastructure in contested environments.[30]

Fuel cells could be applied in several operational ways, depending on power demand and mission duration. One potential use is to integrate fuel cells into environmental control units for expeditionary shelters, providing a clean, quiet alternative to diesel-powered generators that reduces both acoustic and thermal signatures. They could also power deployable command-and-control nodes, data centers, or radar and sensor systems where stable, uninterrupted power is critical. In the mobility space, fuel-cell-powered ground vehicles or unmanned platforms could reduce reliance on petroleum-based fuels, provided that appropriate refueling infrastructure is available or prepositioned. These applications align well with modular ACE concepts, where small, independent teams operate from dispersed locations with limited resupply.

However, there are notable downsides and logistical challenges. Fuel cells—especially hydrogen-based designs—require a secure and consistent fuel supply chain, which may not exist in remote or contested environments. Hydrogen storage and transport introduce both safety considerations (due to its volatility) and infrastructure requirements that are currently limited in most operational theaters. While ammonia-based fuel cells mitigate some of these challenges by using a denser, easier-to-transport fuel, ammonia handling presents its own safety and training demands, and cracking systems add complexity. Maintenance and technical support for these systems also require skilled personnel and spare parts, potentially adding to the operational footprint. As such, fuel cells may be best suited for enduring locations where logistics and safety infrastructure can be established, or for niche deployments where their advantages—low emissions, quiet operation, and integration with renewable energy—outweigh these constraints.

Nuclear Fusion

Nuclear fusion aims to generate heat or electricity by fusing light nuclei, typically deuterium and tritium, to release energy. Two primary approaches are being developed: inertial

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[29] Rami Reshef, "Case Study: GenCell – Fuel Cell Power for Telecoms," GSMA, webpage, May 2022. As of September 15, 2025: https://www.gsma.com/solutions-and-impact/technologies/networks/gsma_resources/case-study-gencell/.

[30] U.S. Pacific Air Forces, *PACAF Strategy 2030: Evolving Airpower*, September 11, 2023.

confinement fusion, which uses lasers to compress thermonuclear fuel, and magnetic confinement fusion, which uses magnetic fields to contain plasma at high pressures.[31] Fusion offers immense energy density—deuterium and tritium fusion produce energy ten million times greater than fossil fuels—and could significantly reduce logistical burdens while replacing fossil fuels as a primary energy source. Deuterium is abundant in water, and tritium can be produced from lithium, making fusion fuels widely available and potentially less expensive than fission fuels. Fusion does not produce radioactive fission products but generates high-energy neutrons that activate materials, creating radioactive byproducts.

Fusion is far more potent than fission, releasing 275 million kilocalories of energy per gram of deuterium,[32] compared with 20 million kilocalories per gram of uranium 235. Recent advancements have accelerated progress in fusion technology, particularly in magnetic confinement systems. Commonwealth Fusion Systems is developing the SPARC tokamak, a compact reactor that uses high-temperature superconducting magnets to confine plasma at 100 million degrees Celsius, enabling net energy gain. The company plans to build the world's first nuclear fusion power plant by 2030, aiming to deliver commercial-scale energy.[33] The SPARC design demonstrates scalability and cost-effectiveness, marking a significant step toward practical fusion energy.

Fusion technologies have been in development for over 60 years, with major projects conducted in organizations like the International Thermonuclear Experimental Reactor (ITER)(magnetic confinement fusion) and Lawrence Livermore National Lab (inertial confinement fusion) leading the field. ITER remains a cornerstone of fusion research, while private companies like Commonwealth Fusion Systems and Helion Energy are accelerating commercialization timelines.[34] Fusion's potential to provide clean, abundant, and reliable energy aligns with global decarbonization goals, offering a pathway to replace fossil fuels and complement renewable energy sources.[35] However, challenges remain, including high costs, complex reactor designs, and the need for skilled personnel. ITER's projected costs exceed $65 billion U.S. dollars, highlighting the financial barriers to widespread adoption.[36] Additionally,

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[31] World Nuclear Association, "Nuclear Fusion Power," webpage, June 5, 2025. As of September 9, 2025: https://world-nuclear.org/information-library/current-and-future-generation/nuclear-fusion-power

[32] Moritz von der Linden, "Fusion Sparks an Energy Revolution," *Wired*, June 19, 2024.

[33] A.J. Creely, M. J. Greenwald, S. B. Ballinger, D. Brunner, J. Canik, J. Doody, T. Fülöp, D. T. Garnier, R. Granetz, and T. K. Gray, "Overview of the SPARC Tokamak," *Journal of Plasma Physics*, Vol. 86, No. 5, 2024

[34] Commonwealth Fusion Systems, "Homepage," undated. As of September 9, 2025: https://cfs.energy

[35] Larua Paddison, "'World's First' Nuclear Fusion Power Plant Planned by Commonwealth Fusion Systems," *CNN*, December 18, 2024.

[36] Maciej Kolaczkowski, "Explainer: Advanced Nuclear Technologies and Their Role in the Energy Transition," World Economic Forum, October 10, 2024. As of September 15, 2025: https://www.weforum.org/stories/2024/10/advanced-nuclear-technologies-energy-transition/

high-energy neutrons produced by fusion reactors could be used to convert U-238 into weapons-grade plutonium, raising proliferation concerns.

Research into fusion technology is a major international undertaking, led by projects like ITER, the world's largest magnetic confinement experiment under construction in southern France, involving 35 nations and expected to begin deuterium–tritium operations between 2035 and 2040 after extensive delays and a multibillion-euro cost overrun.[37] Meanwhile Japan and the United Kingdom are accelerating domestic prototype efforts: Japan is advancing its national fusion power plans, and the United Kingdom is preparing a pilot plant at Culham, supported by record government funding in 2025.[38] Although ITER's goal remains scientific demonstration rather than grid-linked electricity production, project delays and technical hurdles—including sustained plasma confinement, neutron damage to materials, and tritium-breeding challenges—underscore ongoing uncertainties about commercial viability.[39]

Considerations for the USAF for the Employment of Fusion Technology

Global research efforts illustrate that fusion development is widely viewed as strategically important, with significant implications for energy security, decarbonization, and technological leadership.[40] Should compact fusion reactors become feasible for deployment, they could transform expeditionary power logistics for the USAF in enduring locations. However, given current timelines and risks, fusion remains a long-term potential rather than near-term operational capability.

Despite these hurdles, fusion is increasingly viewed as a viable solution for extensive energy security and sustainability. While practical fusion reactors are unlikely to provide power to USAF forces in the near term, ongoing research and development efforts by academic institutions, startups, and international collaborations continue to push the boundaries of this transformative technology.

Portable Nuclear Reactors

Portable nuclear reactors, or microreactors, are of interest for situations where the site power requirements are on the order of 1 MW or greater, and resupply to the site is difficult. In 2021, the upfront cost of nuclear microreactors was comparable to the cost of wind-with-batteries

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[37] ITER Organization, "The Way to New Energy," factsheet, January 2019. As of September 16, 2025: https://www.iter.org/sites/default/files/media/ibf_01_2019.pdf

[38] Nuclear Newswire, "U.K., Japan Step Up Progress Toward Fusion Power Demonstrations," webpage, July 30, 2024. As of September 16, 2025: https://www.ans.org/news/article-6252/uk-japan-step-up-progress-toward-fusion-power-demonstrations/

[39] Jillian Ambrose, "Ministers Pledge Record £410 Million to Support UK Fusion Energy," *The Guardian*, January15, 2025.

[40] Luca Garzotti, "Nuclear Fusion: It's Time for a Reality Check," *The Guardian*, January 22, 2025.

plants. It was estimated that a nuclear microreactor sized between 1 and 5 MW could cost up to $643,000 per MWh per day and would only be suitable for large air bases. This technology has matured, with various manufacturers producing commercially available nuclear microreactors.

Microreactors, often categorized under small modular reactors (SMR) are designed to be compact, scalable, and transportable, making them ideal for remote environments, such as those encountered in ACE operations. The International Atomic Energy Agency defines “small” as under 300 MW. These reactors leverage advanced safety features, such as passive cooling systems, which eliminate the need for external power sources or operator intervention during emergencies. For example, NuScale Power’s modular reactor design, was first approved in January 2023 by the U.S. Nuclear Regulatory Commission and has since produced a new design of 77 MWe in 2025 that was certified in May 2025.[41] This SMR employs an advanced light-water design that relies on natural phenomena like gravity and convection to cool the reactor passively, requiring no supplemental water, external power, or operator intervention.[42] Similarly, GE Hitachi’s BWRX-300 SMR, introduced in 2019, provides a simplified boiling water reactor design that reduces construction and operational costs while maintaining high safety standards.[43] This reactor has recently been deployed in Ontario in 2023 at Darlington Nuclear Generating Station as part of a contract between GE Vernova and Ontario Power Generation, so its application in commercial infrastructure is beginning to mature.

The operational advantages of SMRs extend beyond their compact size. They offer a reliable, carbon-free energy source that can operate continuously for years without refueling, unlike traditional diesel generators, or renewable systems that depend on intermittent energy sources. Companies, such as Nano Nuclear Energy, are exploring SMR designs that emphasize portability and rapid deployment, which could be critical for military applications. These reactors are designed to be transported via standard shipping containers, enabling quick setup in remote locations. Although Nano Nuclear currently has several designs available, including the KRONOS, ZEUS, ODIN, and LOKI SMRs, these reactors are still within the licensing pathway by the Nuclear Regulatory Commission. However, in September 2024 the U.S. Department of Energy granted Nano Nuclear Energy the Gateway for Accelerated Innovation in Nuclear

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[41] Office of Nuclear Energy, “NRC Approves NuScale Power’s Uprated Small Modular Reactor Design,” U.S. Department of Energy, May 30, 2025.

[42] NuScale Power, “NuScale Power Module,” webpage, undated. As of September 9, 2025: https://www.nuscalepower.com/products/nuscale-power-module

[43] GE Vernova, “BWRX-300 Small Modular Reactor,” webpage, undated. As of September 9, 2025: https://www.gevernova.com/nuclear/carbon-free-power/bwrx-300-small-modular-reactor?gad_source=1&gad_campaignid=22426186493&gbraid=0AAAAA_SLd8Sjpxad42ROYL5FunBRcEkMf&gclid=EAIaIQobChMIxruexZ7MjwMVDVpHAR2rHQ0iEAAYASAAEgLYEfD_BwE

voucher award[44] in collaboration with Idaho National Laboratory to support the ZEUS design, which shows that progress is being made to further enhance the portability of SMRs.

Global interest in SMRs has led to significant advancements in their commercialization. According to the OECD Nuclear Energy Agency's SMR dashboard,[45] over 70 SMR designs are currently under development worldwide, with several nearing development, anticipating operations at domestic military installations by 2028.[46] These reactors are being tailored for diverse applications, including off-grid power generation, industrial processes, and military operations. For instance, the World Nuclear Association highlights the potential of SMRs to provide resilient energy solutions in disaster-prone areas or regions with limited infrastructure capabilities closely aligned with ACE's demand for resilient, expeditionary power generation.

Considerations for the USAF for the Employment of Portable Nuclear Reactors

Although the upfront costs of microreactors remain high, their long-term operational benefits—such as reduced fuel costs, minimal maintenance, and enhanced energy security—make them an attractive option for large air bases and forward-operating locations soon. As technology continues to evolve and move beyond the stage of just commercial demonstration, the integration of microreactors into ACE operations could provide PACAF with a robust and sustainable energy solution, ensuring mission readiness in contested environments.

However, deployment of portable nuclear reactors presents additional challenges beyond engineering and cost. Host-nation approval is likely to be a significant obstacle. Many nations have legal, regulatory, and political constraints on nuclear energy infrastructure, particularly mobile nuclear systems operated by foreign military forces. Negotiating basing rights for reactors may require bilateral agreements, explicit liability frameworks, and compliance with international nuclear safety and nonproliferation obligations, which could slow or prevent deployment. Additionally, the USAF currently has limited expertise in operating nuclear power systems for electricity generation; new training pipelines, nuclear-qualified maintenance teams, and dedicated operational procedures would be required. These reactors could also become strategic targets, either physically or via cyber intrusion, given their symbolic and operational value. Physical security, layered defenses, and contingency planning for reactor shutdown or sabotage would need to be incorporated into ACE basing concepts. Finally, reactor end-of-life considerations—including removal, decommissioning, and fuel disposition—would require careful planning to avoid leaving host nations with radioactive materials or environmental

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[44] Gateway for Accelerated Innovation in Nuclear, "Voucher Aware Recipient: Independent Assessment of a Novel Heat Exchanger Concept for Open-Air Brayton Cycle, NE-24-34981," webpage, undated. As of September 9, 2025: https://gain.inl.gov/ne-24-34981/

[45] Nuclear Energy Agency, "NEA Small Modular Reactor (SMR) Dashboard," webpage, undated. As of September 9, 2025: https://www.oecd-nea.org/jcms/pl_73678/nea-small-modular-reactor-smr-dashboard

[46] Todd South, "Army to Lead Nuclear Microreactor Development to Power Bases," *Army Times*, June 4, 2025.

liabilities. These factors suggest that while microreactors hold potential for energy resilience, their integration into ACE concepts of operation requires diplomatic, organizational, and security solutions beyond the technical maturity of the reactors themselves.

Batteries

Battery technology has made major advancements in the past five years, particularly in areas such as energy density, charging speed, and safety. Batteries are an integral part of energy storage and distribution, especially in off-grid applications where power is needed at various stages of an operation. Former director of the energy portfolio at the Defense Innovation Unit, Andrew Higier has suggested that DOW should invest domestically in higher-density, longer lasting batteries, which could improve the cost and efficacy of technologies such as solar and wind powered systems stating that "advanced batteries are the single-greatest cost and a bottleneck for electric platforms because of supply chain and integration issues."[47] With this being an ongoing challenge for domestic manufacturers, progress has been made in improving battery technologies locally. Funding has focused recently on the development of advanced battery technologies, primarily lithium-based batteries, which has remained the standard for research and development focus. Alternatively, other chemistries are being explored in solid-state technology as well as sodium-ion batteries which have incrementally progressed.

In 2021, the DOW invested \$1.25 billion in contract awards to four battery companies.[48] One of those companies was Bren-Tronics, which at that time constructed a Hybrid Energy Advanced Trailer System, a small military trailer that included lithium iron phosphate batteries processing around 14.2 kWh of energy per day, along with other portable battery systems to support the warfighter in wearable technology. As of December 2023, it was reported that the company received additional support from the DIU FAStBat award to standardize lighter, safer, and longer-life batteries for dismounted warfighters—military personnel operating on foot, independent of vehicles, and carrying all essential equipment.[49] Other recipients included Enersys, who developed a commercial prismatic lithium-ion cell and a proven DOW Space 18650 cell design which was used to prototype a high-energy Li6T battery.[50] These advanced

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[47] Halimah Najieb-Locke, Lee Robinson, and Devon Bistarkey, "Collaboration and Standardization Are Key to DoD's Battery Strategy, Meeting U.S. Energy Objectives." Office of the Under Secretary of Defense for Acquisition & Sustainment, June 5, 2023. As of September 16, 2025: https://www.acq.osd.mil/news/office-news/eie/2023/collaboration-and-standardization-are-key-to-dods-battery-strategy-meeting-us-e.html

[48] Shannon Cuthrell, "U.S. Government Taps Four Manufacturers for Wearable Battery Contract," EE Power, undated. May 30, 2021.

[49] Defense Innovation Unit, "Department of Defense to Prototype Commercial Lithium Batteries for Soldier Power, Aviation, and Ground Vehicles," webpage, December 7, 2023. As of September 15, 2025: https://www.diu.mil/latest/department-of-defense-to-prototype-commercial-lithium-batteries-for-soldier

[50] Jake Kauffman, "EnerSys Selected to Develop Lithium-Ion Cell Gigafactory," Defense and Munitions, September 23, 2024.

lithium-ion batteries, especially the BT-70939M, is a MIL-PRF 32565 compliant battery and offers capacities up to 816 Ampere-hours (Ah) with variants like their Nexsys Li6T achieving capacities of up to 1110 Ah.[51]

Currently fielded battery systems across U.S. military operations—including those relevant to PACAF—include ruggedized, mission-tested lithium-ion and lithium iron phosphate units designed for tactical reliability. One such system is the Bren-Tronics F4850, a 5 kWh $LiFePO_4$ battery integrated into hybrid trailer and microgrid configurations with onboard protections against electrical and thermal anomalies.[52] These batteries support high cold-cranking currents—up to 1,100 amps at minus 18 degrees Celsius—making them effective in extreme conditions. These systems are already in service supporting vehicle power, silent watch operations, portable command systems, and energy storage nodes, enabling sustained operations in environments where fuel logistics are limited or risky. Their reliability and performance under harsh conditions make them critical components of current power architecture in distributed and expeditionary missions.

Although advanced battery technologies offer significant operational advantages, their integration is not without challenges. Lithium-based systems, even in safer chemistries like lithium iron phosphate or solid-state designs, are classified as hazardous materials for air transport and require strict packaging, handling, and labeling standards—similar to, and often more stringent than, civilian Transportation Security Administration and International Air Transport Association regulations. These batteries also carry a small but notable risk of overheating or thermal runaway if damaged, as well as supply chain vulnerabilities tied to critical materials like lithium and cobalt. In addition, the adoption of these systems demands new training, safety protocols, and end-of-life recycling processes, all of which increase logistical complexity and cost. Addressing these concerns is essential to ensure that the benefits of advanced batteries—greater energy density, faster charging, and improved safety—translate into practical, reliable support for dispersed and agile operations.

Considerations for the USAF for the Employment of Advanced Batteries

PACAF's emphasis on resilient logistics and operations in austere environments may pave the way for future adoption of advanced battery systems to enhance power provision during ACE operations. Advanced batteries, such as lithium iron phosphate and solid-state designs, provide higher energy density, longer lifespans, and improved safety, making them ideal for supporting operations in remote and contested environments. These technologies enable faster charging and more efficient energy storage, reducing reliance on traditional power infrastructure and

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[51] EnerSys, "NexSys® iON Batteries," webpage, undated. As of September 16, 2025: https://www.enersys.com/en/products/batteries/nexsys/nexsys-ion/

[52] Bren-Tronics, "BT-70980A-50 Series", webpage, undated. As of September 16, 2025: https://www.bren-tronics.com/bt-70980a-50-series.html

enhancing operational flexibility. The rapid adoption of these advancements is critical, as they directly address the logistical challenges of sustaining dispersed forces while ensuring reliable power for mission-critical systems. By prioritizing the integration of these technologies, PACAF can maintain a competitive edge and ensure energy resilience in future operations.

Ultracapacitors

Ultracapacitors—also referred to as supercapacitors—have become increasingly prominent in energy systems because of their ability to deliver rapid bursts of power, withstand extreme environmental conditions, and maintain functionality over more than one million charge and discharge cycles.[53] Unlike batteries, which rely on chemical energy conversion, ultracapacitors store energy electrostatically, enabling near-instantaneous power delivery with low internal resistance and high reliability. These properties make ultracapacitors ideal for applications involving transient loads, backup power, and cold-start operations. Their compact size, rapid recharging time (often under 10 seconds for full charge at rated voltage),[54] and broad operating temperature range (typically minus 40 degrees Celsius to 70 degrees Celsius)[55] offer key advantages in expeditionary and contested environments where logistical agility and equipment uptime are mission critical.

Over the past five years, ultracapacitor technology has progressed significantly through innovations in electrode materials and hybrid device architectures. Carbon-based electrodes such as graphene, activated carbon, or carbon nanotubes have been shown in multiple studies to achieve energy densities in the range of around 10–25 Wh/kg, while sustaining power densities exceeding 10 kW/kg under various hybrid or asymmetric supercapacitor configurations. These characteristics combine high energy storage with rapid power delivery, valuable for applications needing both burst-power and endurance.[56] Hybrid ultracapacitors now incorporate pseudocapacitive materials—such as metal oxides or conductive polymers—to bridge the energy gap between capacitors and lithium-ion batteries, improving energy storage while preserving fast

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[53] Mohd Arif Dar, S. R. Majid, M. Satgunam, C. Siva, Shaheer Ansari, P. Arularasan, and S. Rafi Ahamed, "Advancements in Supercapacitor Electrodes and Perspectives for Future Energy Storage Technologies," *International Journal of Hydrogen Energy,* Vol. 70, June 12, 2024.

[54] Defense Intelligence Agency, *Ultracapacitors as Energy and Power Storage Devices for Commercial and Military Applications*, Defense Intelligence Reference Document, November 1, 2010.

[55] Port Equipment Manufacturers Association, *Ultracapacitors in Port Applications*, 2023. As of September 14, 2025: https://www.pema.org/wp-content/uploads/2023/01/PEMA_Ultracapacitors_Digital_AW.pdf

[56] Zhang, Chao, Yunlong Zhu, Wei Huang, and Haixia Zhang, "Battery-Like Supercapacitors from Vertically Aligned Carbon Nanofibers Coated Diamond," arXiv.org, January 2018. As of September 16, 2025: https://arxiv.org/abs/1801.09763

response times.[57] These developments are coupled with improvements in electrolyte chemistry, reducing Equivalent Series Resistance and enabling thermal stability up to 85 degrees Celsius.[58] Some commercial military-grade supercapacitor modules, such as Skeleton Technologies' SkelStart series, deliver up to 1360 amps of peak current, operate at 24 to 32 volts of direct current, and recharge in under 15 seconds.[59] Such specifications are essential for systems requiring short-duration power with minimal downtime, particularly in high-demand operational cycles.

Currently, ultracapacitors are deployed in military systems as power buffers, cold-cranking supports, and backup power modules. Their high-power density allows them to absorb voltage fluctuations and deliver sharp power bursts without degrading over time. In vehicle platforms, they support silent engine starts even in sub-zero conditions and extend battery lifespan by absorbing regenerative braking energy or mitigating start-up spikes.[60] In radar and communication systems, ultracapacitors maintain voltage stability during transient loads and prevent equipment brownout.[61] For avionics and intelligence, surveillance, and reconnaissance payloads, they act as bridge power during switchover or brief outages, providing 2 to 10 seconds of uninterrupted power at output currents of 200 to 500 amps, depending on system scale.[62] These systems are increasingly modular, designed with stackable cylindrical or prismatic cells, and often include integrated CAN-bus diagnostics for health monitoring.[63]

Considerations for the USAF for the Employment of Ultracapacitors

For PACAF operations across the Indo-Pacific, ultracapacitors offer a powerful tool for improving energy resilience in distributed and austere operating environments. The adoption of

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[57] Himanshu Gupta, Manoj Kumar, Debasish Sarkar, and Prashanth W. Menezes, "Recent technological advances in designing electrodes and electrolytes for efficient zinc ion hybrid supercapacitors," Energy Advances, Issue 9, 2023. As of September 14, 2025: https://pubs.rsc.org/en/content/articlelanding/2023/ya/d3ya00259d

[58] Maitri Libber, Narendra Gariya, and Manoj Kumar, "A Comprehensive Analysis of Supercapacitors with Current Limitations and Emerging Trends in Research," *Journal of Solid State Electrochemistry*, Vol. 29, No. 2, October 14, 2024. As of September 14, 2025: https://link.springer.com/article/10.1007/s10008-024-06107-x

[59] Skeleton Technologies, "SkelStart Ultracapacitor Based Module for Engine Starting," webpage, undated. As of September 14, 2025: https://www.skeletontech.com/skelstart-ultracapacitor-engine-start-module

[60] U.S. Army Engineer Research and Development Center, Cold Start Testing of Ultracapacitor Engine Start Modules on Stryker Vehicles, 2019. As of September 14, 2025: https://apps.dtic.mil/sti/pdfs/AD1079512.pdf

[61] Rajesh Uppal, "Super Capacitors: Powering Innovations from Electric Transport to Military Directed Energy Weapons," Institute for Defence Studies & Technology, August 5, 2023. As of September 14, 2025: https://idstch.com/technology/electronics/super-capacitors-powering-innovations-from-electric-transport-to-military-directed-energy-weapons

[62] Eaton Corporation, "Supercapacitor Modules Resource Center," technical application note, 2018. As of September 14, 2025: https://www.eaton.com/us/en-us/products/electronic-components/topics/supercapacitor-modules-resource-center.html

[63] Maxwell Technologies, "Ultracapacitor Modules (UCAP Series)," product datasheet, 2022. As of September 14, 2025: https://www.maxwell.com/products/ultracapacitors/modules

ultracapacitor modules in power trailers, vehicle fleets, and expeditionary base kits aligns with PACAF's ACE objectives by reducing reliance on fuel-intensive generators and supporting rapid deployment of autonomous and communications infrastructure. Their ability to deliver high peak power while requiring minimal maintenance makes them ideal for forward-operating locations with intermittent or renewable energy sources. For example, ultracapacitors can cold-start vehicle fleets stationed in low-temperature island bases or stabilize solar-charged microgrids used to power mobile radar arrays. Their extended cycle life (1 to 2 million cycles) and lack of chemical degradation make them well-suited for systems that require long-term standby readiness with little logistical burden.[64] By integrating these components alongside existing lithium battery systems, PACAF can ensure more robust, flexible, and fuel-independent power architectures—critical for sustaining combat operations in contested airspace.

Hydrogen Energy Storage Methods

Significant progress in hydrogen storage methods has occurred between 2019 and 2024. In 2021, composite-wrapped Type IV cylinders developed that enabled 700-bar compressed storage systems with volumetric efficiencies of up to 40 kilograms hydrogen per cubic meter (kg/m^3).[65] These tanks use high-strength carbon fiber to reduce weight while maintaining pressure integrity. In 2022, cryo-compressed hydrogen tanks—cooling gas to approximate 20 degrees Kelvin and simultaneously pressurizing it to 350 bar—entered defense-sector pilot programs. These hybrid tanks offered near-liquid hydrogen density (approximately 70 kg/m^3) and significantly reduced boil-off, making them more practical for long-duration missions. Additionally, solid-state technologies have advanced with research on metal hydrides like MgH_2 and $NaAlH_4$ reaching 7 to 11 weight percent hydrogen storage, driven by material breakthroughs in nano structuring and thermal conductivity.[66] Despite these advances, most hydride systems still require high desorption temperatures (200 to 400 degrees Celsius), limiting immediate deployment potential.[67]

Currently, compressed hydrogen remains the most field-ready solution and has been used in expeditionary fuel cell power systems since at least 2020. These systems rely on 350 to 700 bar

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[64] U.S. Department of Energy, *Technology Strategy Assessment: Findings from Storage Innovations 2030: Supercapacitors*, July 2023. As of September 14, 2025: https://www.energy.gov/sites/default/files/2024-02/10_Technology%20Strategy%20Assessment%20-%20%2310%20Supercapacitors_508_rev.pdf

[65] Nicholas A. Klymyshyn, Kriston Brooks, and Nathan Barrett, "Methods for Estimating Hydrogen Fuel Tank Characteristics," *Journal of Pressure Vessel Technology*, Vol. 146, No. 1, February 2024.

[66] Yaohui Xu, Yang Zhou, Yuting Li, Yechen Hao, Pingkeng Wu, and Zhao Ding, "Magnesium-Based Hydrogen Storage Alloys: Advances, Strategies, and Future Outlook for Clean Energy Applications," *Molecules*, Vol. 29, No. 11, 2024. As of September 14, 2025: https://pmc.ncbi.nlm.nih.gov/articles/PMC11173447/

[67] Jianjun Liu, Yong Ma, Jinggang Yang, Lei Sun, Dongliang Guo, and Peng Xiao, "Recent Advance of Metal Borohydrides for Hydrogen Storage," *Frontiers in Chemistry*, Vol. 10, August 16, 2022. As of September 14, 2025: https://pmc.ncbi.nlm.nih.gov/articles/PMC9428915/

cylinders to support mobile microgrids, communications gear, and auxiliary platforms, offering rapid refill cycles and long shelf life.[68] Cryogenic liquid hydrogen (LH_2) tanks are being tested in USAF trials in UAV refueling stations and mobile trailers where endurance and energy density are critical.[69] These systems require multi-layer insulation to limit evaporative losses but provide high-capacity energy per unit volume. In 2023, small-scale metal hydride cartridges began being deployed in remote energy systems, offering low-pressure, ambient-temperature hydrogen storage with no risk of combustion or boil-off.[70] These are particularly attractive for sensors or base-resilience nodes that require periodic bursts of autonomous power.

Considerations for the USAF for the Employment of Hydrogen Energy Storage

Hydrogen continues to gain traction as a tactical energy carrier within the USAF because of its high specific energy (~120 MJ/kg), zero-carbon combustion profile, and compatibility with fuel cell systems. For ACE operations in the Pacific, hydrogen offers a compelling path toward energy independence, particularly in distributed and contested environments. Compressed gas systems can be containerized and pre-positioned via sealift or airlift to forward operating sites, where they support plug-and-play deployment of PEM fuel cell power kits. Cryo-compressed tanks, while more infrastructure-intensive, offer high-density, long-duration storage ideal for UAV ground control and radar systems. Solid-state hydride units—though not yet widespread, present a promising option for prepositioned hydrogen reserves in island installations where logistical resupply may be unpredictable. These can remain dormant until needed, activating on demand with thermal input. By integrating technologies developed between 2020 and 2024 into its logistics and power architecture, PACAF can ensure energy resilience, reduce dependency on fuel convoys, and enhance operational flexibility in distributed, high-tempo environments.[71] Moreover, as the USAF explores alternatives to traditional JP-8 fuel chains, reliable hydrogen storage can become a key enabler of expeditionary energy strategies soon.[72]

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[68] U.S. Army Engineer Research and Development Center–Construction Engineering Research Laboratory, "Army Engineers Demonstrate First Hydrogen Nanogrid at White Sands Missile Range," news release, March 6, 2025. As of September 14, 2025: https://www.erdc.usace.army.mil/Media/News-Stories/Article/4051531/army-engineers-demonstrate-first-hydrogen-nanogrid-at-white-sands-missile-range

[69] Air Force Research Laboratory, "Cryogenic Hydrogen Storage and Refueling Concepts for Unmanned Aerial Vehicles," presented at the AIAA Aviation Forum, American Institute of Aeronautics and Astronautics, 2022. As of September 14, 2025: https://arc.aiaa.org

[70] Clinton Anderson, "Hydrogen Storage Methods," blog post, Fuel Cell Store by, February 12, 2025. As of September 14, 2025: https://www.fuelcellstore.com/blog-section/component-information/hydrogen-storage-information/hydrogen-storage-methods

[71] Under Secretary of Defense for Acquisition and Sustainment, *Department of Defense Operational Energy Strategy*, May 2023. As of September 14, 2025: https://climateandsecurity.org/wp-content/uploads/2024/11/2023-Operational-Energy-Strategy.pdf

[72] Adam Kay, "Fueling Defense: The Future of the United States Air Force," American Gas Association, June 30, 2025. As of September 14, 2025: https://www.aga.org/fueling-defense-the-future-of-the-united-states-airforce/

However, its practical use hinges on how efficiently and safely it can be stored, transported, and deployed under operational conditions. It requires substantial infrastructure for production, compression, and storage, which is challenging to deploy rapidly in contested environments. On-site generation demands significant water and power resources, while transporting hydrogen as compressed gas or liquid adds logistical complexity and regulatory hurdles. Safety risks from leaks and flammability necessitate specialized training and equipment, increasing deployment burdens. Moreover, hydrogen systems are less mature and reliable for rugged, mobile military use compared to batteries or conventional fuels, and their high capital and maintenance costs further limit suitability for rapidly shifting operations.

Cost is also an important factor to consider for hydrogen energy storage. In 2023, researchers at the Department of Energy estimated the cost for hydrogen storage and identified cost drivers. The analysis focused on onboard cryogenic and compressed hydrogen storage for Class 8 Long Haul trucks, large-scale LH_2 storage systems at city gate and trade terminals, and utility-scale engineered underground storage. The Department of Energy found that the typical price to store hydrogen costs $6.50 per kg. Assuming 24 hours of operation, $60/hr for maintenance, and electricity costs of $0.06/kWh, the estimated levelized cost of storage ranged from $4.57 to $67/MWh based on usage.[73] Another recent study in 2024 reported a similar levelized cost of storage of hydrogen fuel cells powered by hydrogen ranging from $62/MWh to $484/MWh illustrating the high cost to implementation of this technology.[74]

Vehicle-to-Grid Systems

Vehicle-to-Grid (V2G) technology enables bidirectional power flow between electrical vehicles (EVs) and the electrical grid, allowing parked vehicles to act as distributed energy storage. By providing instantaneous grid support—like peak shaving, frequency regulation, and resilience during outages, V2G offers a compelling way for military installations to potentially leverage EVs as mobile power reserves. These capabilities align well with PACAF's distributed operations model, offering both energy redundancy and demand-side flexibility in forward-deployed environments.[75]

Since 2020, V2G systems have incorporated smarter bidirectional chargers, enhanced aggregation platforms, and real-time grid services. In 2023, a review reported V2G systems could reduce peak grid demand by up to 20 percent and cut utility costs by 10 to 15 percent,

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73 Cassidy Houchins, Jacob H. Prosser, Max Graham, Zachary Watts, and Brian D. James, "Hydrogen Storage Cost Analysis," briefing, Strategic Analysis, Incorporated, June 2023.

74 Joshua Schmitt, Bikram Roychowdhury, Adam Swanger, Marcel Otto, Jayanta Kapat, "Techno-Economic Analysis of Green Hydrogen Energy Storage in a Cryogenic Flux Capacitor," Proceedings of the ASME Turbo Expo 2024: Turbomachinery Technical Conference and Exposition, Paper No. GT2024-129208 (V006T09A015), August 28, 2024.

75 go-e Magazine, "From Charging to Sharing: What You Need to Know About V2G in 2025," webpage, June 25, 2025. As of September 14, 2025: https://www.go-e.com/en/magazine/vehicle-to-grid

driven by improved inverter efficiency and software aggregators.[76] Nissan plans to launch a commercial bidirectional-capable EV in 2026, marking the first affordable V2G-compatible passenger vehicle.[77] The U.S. Department of Energy's 2025 Vehicle Grid Integration Assessment recognizes V2G's viability, citing over 1.2 million EVs sold in the United States in 2023 as a foundational resource for aggregation.[78] The National Renewable Energy Laboratory projects significant grid support from standardizing V2G charge protocols and integrating EVs as controllable loads.[79]

Globally, utility and military pilots are operational. For example, go-e (a German firm) offers bidirectional charging units supporting up to 11 kW power flows, suitable for residential or depot installations.[80] A pilot at Los Angeles Air Force Base demonstrated an all-EV fleet providing more than 500 kW to base facilities—enough to power 50 homes for 3.5 hours.[81] Department of Energy models confirm that EVs, acting as aggregations, can perform frequency regulation, demand response, and spinning reserve—unlocking grid services during high-stress periods.[82] Recent papers document V2G's controlled discharge capabilities and advanced learning algorithms for grid dispatch.[83]

Considerations for the USAF for the Employment of Vehicle-to-grid Technology

For PACAF, integrating V2G into base energy architecture presents an immediate opportunity. Base EV fleets—assigned to command, security, and utility missions—can serve

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[76] Guozhong Chen and Zhaoyun Zhang, "Control Strategies, Economic Benefits, and Challenges of Vehicle-to-Grid Applications: Recent Trends Research," *World Electric Vehicle Journal*, Vol. 15, No. 5, March 15, 2024. As of September 15, 2025: https://www.mdpi.com/2032-6653/15/5/190

[77] Nissan Motor Corporation, "Nissan to Launch Affordable Vehicle-to-Grid Technology in 2026," news release, October 10, 2024. As of September 15, 2025: https://global.nissannews.com/en/releases/nissan-to-launch-affordable-vehicle-to-grid-technology-in-2026

[78] U.S. Department of Energy, *Vehicles-to-Grid Integration Assessment Report*, January 2025. As of September 15, 2025: https://www.energy.gov/sites/default/files/2025-01/Vehicle_Grid_Integration_Asseessment_Report_01162025.pdf

[79] National Renewable Energy Laboratory, "Electric Vehicle Grid Integration," webpage, 2023. As of September 15, 2025: https://www.nrel.gov/transportation/project-ev-grid-integration

[80] Marc Escoto, Antoni Guerrero, Elnaz Ghorbani, and Angel A. Juan, "Optimization Challenges in Vehicle-to-Grid (V2G) Systems and Artificial Intelligence Solving Methods," *Applied Sciences*, Vol. 14, No. 12, June 15, 2024. As of September 15, 2025: https://www.mdpi.com/2076-3417/14/12/5211

[81] California Energy Commission, "Los Angeles Air Force Base Vehicle-to-Grid Pilot Project," project summary, 2018. As of September 15, 2025: https://www.energy.ca.gov/programs-and-topics/programs/clean-transportation-program/vehicle-grid-integration/laafb-v2g-pilot

[82] Lotfi, Siamak, Mostafa Sedighizadeh, Rezvan Abbasi, and Seyed Hossein Hosseinian, "Vehicle-to-grid bidding for regulation and spinning reserve markets: A robust optimal coordinated charging approach," Energy Reports, Vol. 11, 2024. As of September 16, 2025: https://doi.org/10.1016/j.egyr.2023.12.044

[83] Songling Pang, Kaidi Fan, Meiyi Huo, "Charge and Discharge Scheduling Method for Large-Scale Electric Vehicles in V2G Mode via MLGCSO," *Scientific Reports*, May 9, 2025. As of September 15, 2025: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12064748

dual roles as transport and mobile energy assets. V2G chargers at forward operating locations would enable bidirectional energy flows during peak demand or outages, reducing reliance on fuel-based generators and enhancing microgrid resilience. As Nissan's 2026 V2G-capable vehicles become available, the cost of enabling this capability will hinge largely on the supporting infrastructure. Rough-order-of-magnitude estimates for bidirectional V2G chargers are between $2,500 to $12,000 each, with installation adding $600 to $12,700 depending on site complexity. Expressed in terms of storage capacity, these systems generally fall between $200 and $400 per kWh, with a levelized cost of storage between $0.085 to $0.243/kWh.[84] For PACAF, this means that transitioning from a pilot program to operational use could be achieved at a modest capital outlay compared with fixed energy storage systems, while leveraging the EV fleet as both transport and a mobile, resilient energy reserve.

Aggregated EV power across a squadron could provide hundreds of kilowatts of reserve power, usable for comms nodes, airfield lighting, or medical support. With minimal infrastructure investment beyond chargers and control systems, V2G enables PACAF to leverage existing resources for agile, clean, and redundant energy support in distributed Pacific operations.

Rough Order of Magnitude Costs for Power Technologies

As mentioned in previous sections, a rough magnitude of cost can be estimated to anticipate a potential cost for emerging power and water technologies. Using academic and industry literature along with vendor interviews, we were able to extract hypothetical ranges for costs associated with initial purchase, operations, and field maintenance, which are accounted for in the technology's levelized cost of energy (LCOE).

LCOE is a critical metric for long-term planning and acquisition decisions, especially when considering emerging technologies to sustain distributed operations. The LCOE represents the average cost of producing one megawatt-hour (MWh) of electricity over the lifetime of a system, accounting for both capital gain and operating expenses. To balance energy resilience and expeditionary requirements, LCOE is vital in determining whether a technology such as airborne wind turbines, hydrogen energy storage, or portable nuclear reactors can be viably scaled for sustained operations.

The general formula for calculating LCOE is:

[84] Jingxuan Geng, Bo Bai, Han Hao, Xin Sun, Ming Liu, Zongwei Liu, Fuquan Zhao, "Assessment of Vehicle-Side Costs and Profits of Providing Active Services with Electric Vehicles," eTransportation, Vol. 19, 2024. As of September 15, 2025: https://www.sciencedirect.com/science/article/pii/S2590116823000784

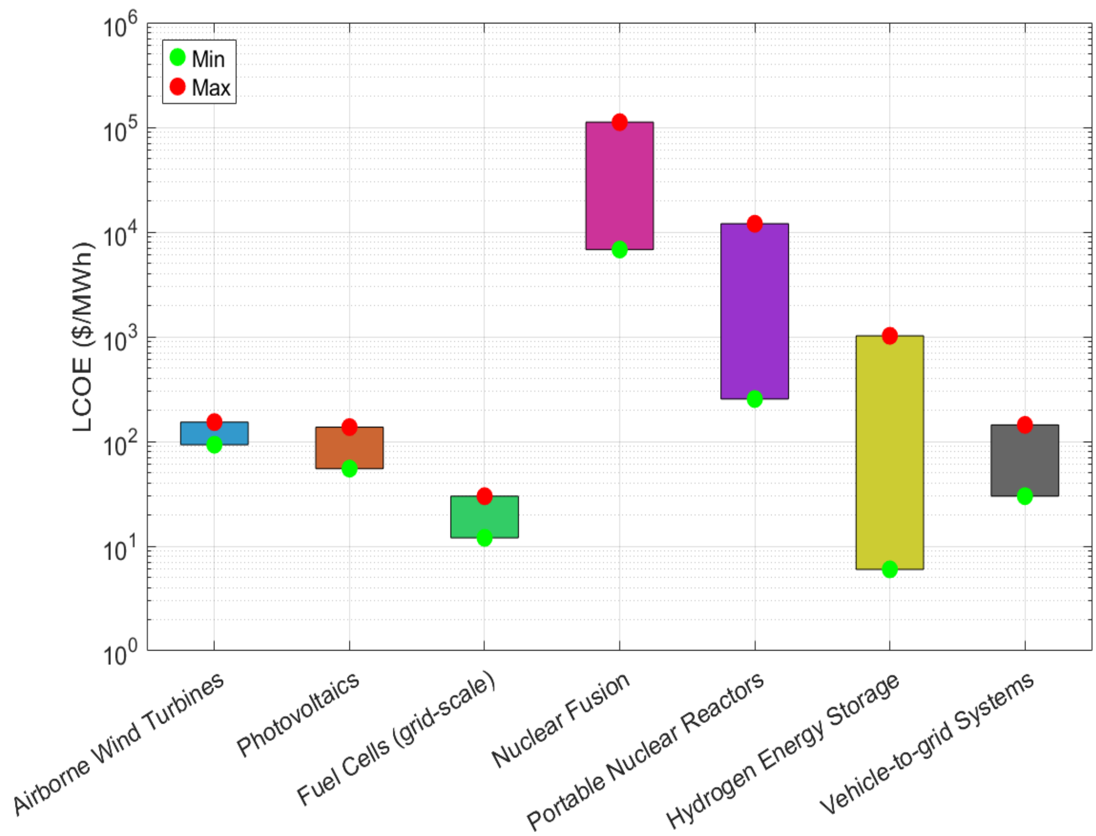
$$LCOE = \frac{\sum_{t=1}^{N} \frac{I_t + O_t + F_t}{(1+r)^t}}{\sum_{t=1}^{N} \frac{E_t}{(1+r)^t}}$$

Where I_t is the initial investment expenditure in year t, O_t is the operation and maintenance costs in year t, F_t is the fuel costs in year t (if applicable), E_t is the electricity generation in MWh in year t, r is the discount rate (assumption = 0) and N is the system lifetime in years.

An estimated lifetime of 10 years for power technologies was used as a baseline to assume potential costs, as it balances realistic operational durability with the uncertainty of field conditions. For photovoltaics, fuel cells, airborne wind turbines, hydrogen storage, and vehicle-to-grid systems published lifetimes often extend beyond a decade, but actual deployment in expeditionary or high-tempo environments may accelerate wear.[85] Using 10 years as baseline therefore provides a conservative yet analytically sound horizon for cost comparison. Portable nuclear reactors and fusion systems, by contrast, require exclusion from this simplification since their design lives, regulatory pathways, and technology readiness levels are fundamentally different and not yet directly comparable to more mature, commercially available systems. Hence, portable nuclear reactors and fusion systems' lifetime was estimated as 30 years for this study. To understand what the potential investment opportunity may look like in relation to each technology, the LCOE is illustrated in Figure 2.1. LCOE values mentioned thus far were scaled to reflect the cost in $/MWh. Storage technologies such as batteries and ultracapacitors are omitted, excluding hydrogen energy storage provided production is available. For further understanding of the breakdown of the LCOE values for the estimated costs, operation and maintenance (O&M), and estimated lifetime output refer to Table 2.1.

[85] U.S. Department of Energy, Solar Energy Technologies Office, "End-of-Life Management for Solar Photovoltaics," webpage, undated. As of September 15, 2025: https://www.energy.gov/eere/solar/end-life-management-solar-photovoltaics; U.S. Department of Energy, Hydrogen and Fuel Cell Technologies Office, "DOE Technical Targets for Fuel Cell Backup Power Systems," webpage, undated. As of September 15, 2025: https://www.energy.gov/eere/fuelcells/doe-technical-targets-fuel-cell-backup-power-systems

Figure 2.1. Levelized Cost of Energy Ranges by Technology



As shown in Table 2.1, the least expensive, near-term power solutions are the mature, low-capital assets—airborne wind turbines, photovoltaic arrays, and grid-scale fuel-cell systems—which cost a few hundred thousand dollars to a few million and deliver the lowest levelized cost of energy. Energy-storage options follow a similar pattern: lithium-ion batteries and ultracapacitors are inexpensive to procure but have higher LCOE per stored kWh, while hydrogen-storage systems span a wide cost range and remain less mature. The most capital-intensive options—portable nuclear reactors and nuclear-fusion concepts—require tens of millions to billions of dollars and yield very high LCOE, indicating they are long-term, strategic investments rather than immediate solutions.

Table 2.1. Rough Order of Magnitude Costs for Acquisition, Operations and Maintenance, and LCOE for Emerging Power Technologies

Technology	Estimated Capital Cost	Estimated Lifetime (years)	Annual Output (MWh)	Estimated O&M (10 years)	Estimated O&M (10 years)	Potential Total Cost	Lifetime Output (MWh)	LCOE ($/MWh)	Potential Initial Cost
Airborne Wind Turbines	$1.7 million – $2.8 million	10	2,190	$340,000	$560,000	$2.04 million– $3.36 million	21,900	$93–$200	$220,000
Photovoltaics	$100,000 – $250,000	10	219	$20,000	$50,000	$120,000– $300,000	2,190	$55–$137	$30,000
Fuel Cells (grid-scale)	$80,000 – $200,000	10	788	$16,000	$40,000	$96,000– $240,000	7,880	$12–$30	$24,000
Nuclear Fusion	$1 billion – $25 billion	30	7,884	$600 million	$1.5 billion	$1.6 billion– $26.5 billion	236,520	$6,800– $112,000	$900 million
Portable Nuclear Reactors	$50 million – $3 billion	30	7,884	10,000,000	$60 million	$60 million– $3.0 billion	236,520	$254– $12,930	$500 million
Batteries (Li-ion, grid)	$400 – $900	10	10 (storage only)	$80	$180	$480–$1,080	100	$4.8– $10.8/kWh	$100
Ultracapacitors	$2,000 – $10,000 (large)	10	10 (storage only)	$400	$2,000	$2,400– $12,000	100	$24– $120/kWh	$1,600
Hydrogen Energy Storage	$5,000 – $1 million	10	100	$1,000	$20,000	$6,000–$1.02 million	1,000	$6–$1,020	$19,000
Vehicle-to-grid Systems	$2,500 – $12,000	10	10 (per charger)	$500	$2,400	$3,000– $14,400	100	$30–$144	$1,900

Note: Values expressed in this table are rough order of magnitude estimates and do not reflect the actual cost to market for specific technologies. O&M=operations and maintenance; MWh=megawatt hour
SOURCE: RAND LCOE Analysis
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Chapter 3. Water Filtration and Decontamination Technologies

In this chapter, we describe four technologies we evaluated as alternative methods to providing water during ACE operations. These technologies could be used to provide drinking water for personnel or other needed water for mission functions. We present the same information as we did for energy: a general description of the technology, how the technology has evolved since the 2021 survey, key considerations for USAF employment of the technology, and some ROM costs. Given that the cost to market for emerging water technologies is less substantive than that for power, we omitted a full rough order of magnitude analysis for water. You will find any potential price points for emerging water technologies in-text.

Water Filtration and Decontamination

Clean water access is a mission-critical requirement for PACAF forces operating across the Indo-Pacific, where remote installations and mobile operations often lack reliable municipal sources. Water filtration and decontamination systems must be robust, portable, and capable of addressing microbial, chemical, and particulate contaminants under diverse environmental conditions. Within the USAF's ACE strategy, point-of-need water purification ensures force sustainment without relying on fuel-heavy water convoys or vulnerable supply lines.[86]

Water purification technology has accelerated between 2020 and 2025 with key breakthroughs in filtration materials, process automation, and modular system design. In 2024, scientists at the University of Chicago developed a rapid fabrication technique for polymer membranes, improving throughput and enhancing membrane uniformity for high-efficiency contaminant removal.[87] Oxydus, established in 2021, launched a line of electrochemical oxidation systems that use advanced electrode reactions to disinfect water without added chemicals—ideal for military scenarios with limited consumables.[88] A 2024 study in Environmental Advances by Mamba et al. demonstrated that hybrid ceramic-carbon nanocomposite membranes could remove 99.9 percent of heavy metals and bacteria, all while

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[86] Moer T. Zachary, Chini, Christopher M., "Contested Agile Combat Employment," Air & Space Operations Review, Vol. 1, No. 3, Air University, 2023. As of September 16, 2025: https://www.airuniversity.af.edu/Portals/10/ASOR/Journals/Volume-1_Number-3/Chini_Contested_Agile_Combat_Employment.pdf

[87] Paul Dailing, "UChicago Scientists Invent Faster Method to Make Advanced Membranes for Water Filters," University of Chicago, October 24, 2024. As of September 15, 2025: https://news.uchicago.edu/story/uchicago-scientists-invent-faster-method-make-advanced-membranes-water-filters

[88] Oxydus, homepage, undated. As of June 2025: https://www.oxydus.com

exhibiting low fouling and high water flux.[89] Genesis Water Technologies, with product updates as recent as 2023, introduced compact electrocoagulation and nanofiltration systems tailored for high-salinity and industrial wastewater in decentralized applications.[90]

Traditional methods such as RO, granular activated carbon, and ultraviolet (UV) irradiation remain widely used, as noted by the U.S. EPA, but are increasingly supported by more efficient and resilient nano-enhanced systems.[91] In a 2025 review article published in *Environments*, the authors discuss advances in graphene-based electrochemical and fluorescence sensors , including evidence that graphene oxide membranes could remove a range of pharmaceutical and metal contaminants in under 10 minutes per liter, making them highly suited for field-scale deployment.[92] The International Water Association reported in 2024 that advancements in modular nanotechnology-based filtration allow operation without chemical pre-treatment, enabling usage in mobile and emergency settings.[93] According to BCC Research, more than 60 percent of recent filtration innovations are designed for off-grid or rapid-deployment environments, marking a global pivot toward decentralized treatment approaches.[94] These systems are increasingly durable, self-cleaning, and compatible with solar or generator power, improving their suitability for remote PACAF environments.

Rough-order-of-magnitude costs for advanced nano-enhanced water filtration systems are higher than traditional RO or granular activated carbon setups but offer significantly greater portability and reduced maintenance demands. Publicly available market data suggest that portable, field-deployable nano-filtration units capable of producing 500 to 1,000 liters per day typically range from $5,000 to $20,000, depending on throughput, power integration, and automation features. Larger modular containerized systems designed for mobile, or emergency deployment can range from $50,000 to over $200,000, reflecting higher capacity and integrated renewable power options. While these upfront costs exceed those of conventional systems, the

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[89] Mamba, P.P., et al., "The Removal of Pathogenic Bacteria and Dissolved Organic Matter from Fresh Water Using Microporous Membranes: Insights into Biofilm Formation and Fouling Reversibility," *Bioadhesion and Biofilm Research*, Vol. 40, 2024. As of September 15, 2025: https://www.tandfonline.com/doi/full/10.1080/08927014.2024.2339438

[90] Genesis Water Technologies, "Industrial Water Purification Systems for Heavy Industries," webpage, October 25, 2024. As of September 15, 2025: https://genesiswatertech.com/blog-post/industrial-water-purification-systems-for-heavy-industries-2/

[91] U.S. Environmental Protection Agency, "Overview of Drinking Water Treatment Technologies," webpage, March 28, 2025. As of September 15, 2025: https://www.epa.gov/sdwa/overview-drinking-water-treatment-technologies

[92] Khalajiolyaie, Akram, and Cuiying Jian, "Advances in Graphene-Based Materials for Metal Ion Sensing and Wastewater Treatment: A Review," *Environments*, Vol. 12, No. 2, 2025. As of September 15, 2025: https://www.mdpi.com/2076-3298/12/2/43

[93] Divya Dhamija, "Innovation in Water Filtration: A Global Perspective," BCC Research, blog post, June 5, 2024. As of September 15, 2025: https://blog.bccresearch.com/innovation-in-water-filtration-a-global-perspective

[94] Nambi Katu and Kiu Special, "Water Purification Technologies: Innovations and Challenges," *Research Invention Journal of Engineering and Physical Sciences*, Vol. 4, No. 1, March 2025. As of September 15, 2025: https://www.researchgate.net/publication/390058057_Water_Purification_Technologies_Innovations_and_Challenges

reduced need for chemical pre-treatment, lower consumable usage, and compatibility with solar or generator power can lower lifetime operating costs, an important factor for PACAF's remote and distributed operations.

Considerations for the USAF for the Employment of Water Decontamination and Filtration Technology

For PACAF, the strategic advantage of integrating next-generation water purification systems lies in mobility and independence. Genesis Water Technologies' (2024) modular units can be integrated into trailers or containerized kits for island outposts, while Oxydus' 2021 electrochemical systems reduce the need for filter replacements or chemical inputs. University of Chicago's 2023 membrane technology could further reduce logistical overhead for small-unit filtration, especially in soldier-wearable or portable devices. Given the threat of natural disasters, water scarcity, or contamination in contested environments, these rugged technologies offer PACAF a path toward sustained, in-theater water independence. Integration with existing microgrid or expeditionary systems would ensure hydration, medical sanitation, and equipment decontamination—ensuring mission continuity even when traditional supply lines fail.[95]

Reverse Osmosis

RO is a high-pressure membrane separation process that removes a broad spectrum of contaminants—dissolved salts, heavy metals, organics, and pathogens—with typical rejection rates of 95 to 99 percent. PACAF frequently encounters environments where potable water sources are limited or contaminated. RO, already in use at large-scale installations like the Cape Coral municipal plant since 1977, is a scalable and proven solution for both base-level and expeditionary water purification alike.[96]

Since 2020, both material improvements and anti-fouling techniques have enhanced RO efficiency and durability. In 2024, Zhang et al. demonstrated that incorporating antibacterial nanoparticles into thin-film composite membranes improved salt rejection and significantly reduced biofouling rates.[97] In a similar study, Long et al. introduced graphene oxide–carbon nanotube interlayers that increased water flux by 30 percent while maintaining contaminant

[95] Allyson Park, "Army to Live Off the Land in Contested Environments," *National Defense Magazine*, January 12, 2024. As of September 15, 2025: https://www.nationaldefensemagazine.org/articles/2024/1/12/army-to-live-off-the-land-in-contested-environments

[96] City of Cape Coral Utilities, *2022 Annual Drinking Water Quality Report*, undated. As of September 15, 2025: https://cms4files.revize.com/capecoralfl/2022%20Water%20Quality%20Report%206.22.2023.pdf

[97] L. Zhang, Y. Qian, D. Zhao, "Antifouling and antibacterial bioactive metabolites of marine fungus (terrein)/polyamide thin-film composite reverse osmosis membranes for desalination applications," *Desalination*, Vol. 572, March 2024. As of September 15, 2025: https://www.sciencedirect.com/science/article/abs/pii/S0011916423007725

rejection above 98 percent.[98] Additionally, a 2025 report (citing the Institute of Process Engineering, Chinese Academy of Sciences) noted a novel dual-function RO membrane with sustained antibacterial and anti-adhesion properties, improving anti-fouling performance under high-salinity conditions.[99]

Industrial-scale RO desalination remains widespread, with plants like those detailed by Genesis Water Technologies in 2023 utilizing energy-efficient spiral-wound membranes capable of treating brackish and seawater sources for forward-operating bases.[100] Portable and under-sink systems—such as tankless RO units like the Waterdrop X12 Pro—now include automated total dissolved solids monitoring, auto-shutoff, and capacities of 600 GPD, making them efficient for small units or vehicle-mounted kits.[101] Meanwhile, advanced pretreatment strategies, including intermittent ultraviolet-irradiation and enzymatic cleaning, are being implemented to mitigate biofouling and extend membrane life, as outlined in a 2025 report in *Environmental Science & Technology*.[102]

Small, portable RO units suitable for field operations and producing up to 1,000 liters per day generally cost $1,000 to $5,000, while mid-scale skid-mounted military or disaster-relief units capable of treating 5,000–20,000 liters per day can range from $15,000 to $60,000.[103] Large, containerized RO plants designed for semi-permanent, or base-level use may cost $100,000 to $500,000 or more, particularly when equipped with automated controls, redundancy, and integrated pretreatment systems.[104] Although RO remains one of the most established water purification methods, its higher power demands and need for regular membrane maintenance can

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[98] Li Long, C. Wu, Z. Yang, C. Tang, "Carbon Nanotube Interlayer Enhances Water Permeance and Reduces Fouling for Thin-Film Composite Membranes," *Environmental Science & Technology*, 2022. As of September 15, 2025: https://pubs.acs.org/doi/10.1021/acs.est.1c07332

[99] Mingjiao Lang, Jianquan Luo, Yinhua Wan, Xinyan Wang, Xiangrong Chen, Guangyong Zeng, "Dual-Functional Reverse Osmosis Membranes: A Novel Approach to Combat Biofouling with Enhanced Antibacterial and Antiadhesion Properties," *Journal of Membrane Science*, Vol. 718, March 2025. As of September 15, 2025: https://www.sciencedirect.com/science/article/abs/pii/S0376738825000122

[100] Genesis Water Technologies, "Reverse Osmosis Desalination Solutions," webpage, undated. As of September 15, 2025: https://genesiswatertech.com/solutions/drinking-water/reverse-osmosis-desalination/

[101] Home Depot, "Waterdrop: Tankless 5-in-1 Under-Sink Reverse Osmosis Water Filtration System with 600 GPD Membrane (B-WD-D6-B)," product page, undated. As of September 15, 2025: https://www.homedepot.com/p/Waterdrop-Tankless-5-in-1-Under-Sink-Reverse-Osmosis-Water-Filtration-System-with-600-GPD-Membrane-B-WD-D6-B/319413739

[102] Y. Myshkevych, G. Scarascia, J. Medina, S. Narayanasamy V. Satagopam, P. Hong, "Effectiveness of Combined UV-C and Bacteriophage Approach Over Repeated Cleaning Cycles to Alleviate Membrane Fouling of Anaerobic Bioreactors," *Chemical Engineering Journal Advances*, June 18, 2025. As of September 16, 2025: https://www.sciencedirect.com/science/article/pii/S2666821125000936

[103] Alibaba, "Reverse Osmosis Plant 1000 Liter Day Price," product listing, undated. As of September 16, 2025: https://www.alibaba.com/showroom/reverse-osmosis-plant-1000-liter-day-price.htm

[104] Lenntech, *"Reverse Osmosis Plants / RO Systems,"* webpage, undated. As of August 13, 2025: https://www.lenntech.com/systems/reverse-osmosis/ro/rosmosis.htm

increase operational costs in remote PACAF settings compared with newer low-maintenance technologies.

Considerations for the USAF for the Employment of Reverse Osmosis Technology

For PACAF operations across remote islands and field-forward areas, modern RO systems offer scalable water solutions. Expeditionary RO units—containerized or trailer-mounted—can be pre-positioned with integrated anti-fouling membranes and compact energy recovery devices, minimizing energy demands (typically 3–5 kWh/m^3) and reducing maintenance cycles.[105] At smaller scales, portable units using graphene-enhanced membranes (2023–2025 tech) permit on-demand purification of local water sources with limited power and logistical support. Anti-biofouling advances make systems more resilient in tropical or bio-rich environments, reducing downtime and chemical use. By aligning evolving RO technologies with PACAF water logistics, installations can secure reliable, self-sufficient water sources under ACE models—enhancing both force health and mission endurance.

Capacitive Deionization

Capacitive deionization (CDI) is an electrochemical technique that removes salt and charged contaminants from water by applying a voltage across porous electrodes, causing ions to absorb into electrical double layers. Unlike RO or thermal methods, CDI operates at low pressure, consumes less energy, and can be regenerated for repeated cycles. For PACAF, which operates across remote bases and islands, CDI offers an efficient, modular water purification method suitable for mid-range brackish water use—complementing RO and advanced filtration systems by reducing energy use and simplifying logistics.

Between 2021 and 2025, CDI has evolved through improvements in electrode materials and hybrid architecture. In 2024, Wang et al. published a carbon aerogel CDI cell that achieved salt removal efficiencies of over 80 percent at low voltages, while excelling in cycle durability and fouling resistance.[106] In 2023, Siontech Co., Ltd. continued developing its proprietary Smart Deionization (SDI) line of capacitive deionization systems, which incorporate ion-exchange membranes to improve desalination performance compared with conventional CDI approaches.[107] Patent filings attributed to Siontech describe electrode and module configurations

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[105] Genesis Water Technologies, "Seawater Desalination Energy Recovery Systems: A Detailed Evaluation," blog post, August 15, 2024. As of September 15, 2025: https://genesiswatertech.com/blog-post/seawater-desalination-energy-recovery-systems/

[106] Ying Wang, Yifan Wang, and Yujie Wang, "Aqueous Zinc–Air Batteries: Recent Progress and Perspectives," ACS Sustainable Chemistry & Engineering, 2024; Ying Wang, Yifan Wang, and Yujie Wang, "Aqueous Rechargeable Zinc–Air Batteries: Recent Progress and Perspectives," ACS Materials Letters, 2024.

[107] Siontech Co., Ltd., "Smart Deionization (SDI) Introduction," webpage, undated. As of September 16, 2025: https://www.siontech.com/english/sdi_intro

designed to increase adsorption efficiency and desalination throughput, building toward scalable membrane-CDI (MCDI) platforms.[108] Another study by Lee et al. reported a hybrid CDI/electrodialysis stack that doubled ion-removal rates without increasing power consumption.[109] Recently, Zhang et al. introduced a graphene-enhanced CDI electrode that reduced energy usage by 30 percent per cubic meter of deionized water.[110] These advances collectively aim to make CDI equally effective as RO for brackish water while maintaining lower operational complexity.

CDI units are now commercially available for brackish water desalination, small-community water systems, and industrial water reuse. For example, modular CDI systems from MSE Supplies are configurable, containerized units capable of treating 10–50 m^3/day with energy use as low as 0.8 kWh/m^3.[111] Indian-manufactured CDI purification plants are deployed at construction camps and field offices, offering portable potable water production with minimal infrastructure.[112] Hand et al. validated CDI for removing perchlorate and nitrate, highlighting targeted contaminant removal in brackish sources—an application relevant for mission-critical water chemistry control.[113] These systems are noted for low-pressure operation, zero chemical dosing, and rapid startup—features that align well with field-expedition requirements.

Although there are advantages that come with using capacitive deionization as a method for water treatment, it will be challenging to incorporate this technology into operations. First, the CDI cells are costly, having few vendors that are commercially available. One example is MSE Supplies, a U.S. based company, whose CDI configurable cell comes out to over $13,000 initially, with energy consumption ranging from $0.02–$0.22 per cubic meter, depending on the feed water and operating conditions.[114] The maintenance costs are relatively low, which is one of the technology's primary benefits. The major operating expense is the eventual replacement of the electrodes, which is based on the lifespan, water chemistry exposure, and number of

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108 Justia Patents, "Patents Assigned to Siontech Co., Ltd.," webpage, undated. As of September 16, 2025: https://patents.justia.com/assignee/sion-tech-co-ltd

109 Chen, T.H., Chen Y..Tsai, S., Wang, D., Hou, C., and C.J. Shih, "Development of an Integrated Capacitive–Electrodialysis Deionization System (CapED) for Continuous Low-Energy Electrochemical Deionization", Separation and Purification Technology, Vol. 274, November 2021. As of September 16, 2025: https://www.sciencedirect.com/science/article/abs/pii/S1383586621007735

110 Y. Zhang, et al., "Graphene-Enhanced CDI Electrodes," *ACS Materials Letters*, Vol. 6, No. 3, March 2024.

111 MSE Supplies, "Membrane Capacitive Deionization Systems: Product Specifications," webpage, 2024. As of September 16, 2025: https://www.msesupplies.com

112 IndiaMart, "Capacitive Deionization Water Purifier Plants: Mid-Scale Portable Units," webpage, 2023. As of September 16, 2025: https://www.indiamart.com

113 S. Hand, R. Cusick, "Emerging Investigator Series: Capacitive Deionization for Selective Removal of Nitrate and Perchlorate: Impacts of Ion Selectivity and Operating Constraints on Treatment Costs," *Environmental Science: Water Research & Technology*, No. 4, 2020. As of September 16, 2025: https://pubs.rsc.org/en/content/articlelanding/2020/ew/c9ew01105f

114 MSE Supplies, "Membrane Capacitive Deionization (MCDI) Configurable Cell," webpage, undated. As of September 16, 2025: https://www.msesupplies.com

operational cycles. Another potential disadvantage would be electrode degradation from oxidation or fouling from organic matter. Given that the technology is still maturing, alternative electrode chemistries and support systems are being explored.

Considerations for the USAF for the Employment of Capacitive Deionization Technology

For PACAF, CDI presents a robust complement to existing water technologies. In areas where salinity is moderate (e.g., brackish groundwater or reclaimed wastewater), CDI can be integrated as a pre-treatment stage before RO or used independently for drinking water provision. Portable CDI systems (10–50 m^3/day) can be airlifted or mounted on trailers to support remote bases. Hybrid CDI/electrodialysis units offer expandable capacity for small camps or sensor hubs needing precise ion control. With energy consumption of around 1 kWh/m^3 and no high-pressure pumps or chemical supplies required, CDI systems reduce logistic overhead and vulnerability. Moreover, anti-fouling electrode innovations from 2024 make CDI more resilient in tropical or bio-rich environments—ideal for Indo-Pacific use. By incorporating CDI-generated water into broader purification architectures, PACAF can build layered protection—ensuring redundancy, reducing weight, and enhancing mission readiness through diversified water sources.

Atmospheric Water Harvesting

Access to clean water is a pressing challenge in arid and remote regions, where traditional water infrastructure is often unavailable. Two emerging technologies—atmospheric water harvesting (AWH) and combustion-based water recovery—offer innovative solutions, though for AWH there have been far more significant advancements and practical applicability.

The companies working on research and development of AWH units imply that these units can provide water in almost any environment, drawing resources from fog and dew (primary methodologies) using advanced materials with high absorption capacity. Although still a developmental concept, AWH has steadily transitioned toward commercial viability. Recent breakthroughs, particularly in material science, have significantly improved its efficiency. Metal-organic frameworks now enable water collection in environments with relative humidity as low as 10 percent, making AWH viable even in desert conditions. For instance, in 2024, researchers at the University of Nevada, Las Vegas developed a device capable of producing up to ten liters of water per day using metal-organic frameworks.[115]

Energy efficiency has also improved through hybrid systems combining solar power with advanced heat exchangers, reducing energy consumption and enabling off-grid applications.[116] In

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[115] John Domol, "Watershed Moment: Engineers Invent High-Yield Atmospheric Water Capture Device for Arid Regions," University of Nevada, Las Vegas UNLV News, October 24, 2024.

[116] Joe Rojas-Burke, "Mapping the Way for Harvesting Water from Air," *ASU News*, March 17, 2025.

2020, the Defense Advanced Research Projects Agency established the Atmospheric Water Extraction (AWE) program (AWE is synonymous with AWH), targeting lightweight portable systems for individual drinking water needs (5-7 liters/day) and larger stabilization units capable of supporting company-size groups (~150 personnel).[117] Atlantis Solar's "H2O Elite" line offers commercial atmospheric water generators producing around 20+ liters per day under favorable environmental conditions. While data on performance across broad temperature and humidity ranges is limited, these systems show strong promise for supporting expeditionary water independence.[118]

These advancements have expanded AWH's applications, including disaster relief, urban water supplementation, and agricultural irrigation. Its ability to provide water covertly and reduce reliance on bottled water transport makes it particularly valuable for military and humanitarian operations. Continued investment in research and rigorous testing for reliability is essential to further enhance its scalability and durability.

Combustion-based water recovery captures water from engine exhaust, leveraging the fact that burning one gallon of fuel produces one gallon of water. The U.S. Army has demonstrated prototypes capable of recovering water per gallon of fuel burned, using systems with carbon beds, resin beds, heat exchangers, and condensers.[119] Despite its theoretical promise, this technology faces significant logistical challenges. Systems are large, heavy, and require extensive maintenance, and water yield is limited by combustion stoichiometry.[120] While research has explored ways to improve efficiency, such as advanced filtration and condensation technologies, combustion-based recovery remains less efficient and scalable compared to AWE.[121] Given the critical need for reliable water sources in isolated and water-scarce regions, continued investment in AWE technology is recommended to enhance its reliability, scalability, and integration into operational contexts.

Cost remains a key consideration. Small, portable AWE systems for tactical use currently range from $1,500–$5,000 per unit and produce 5–20 L/day, translating to a relatively high cost

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[117] Defense Advanced Research Projects Agency, "AWE: Atmospheric Water Extraction," webpage, undated. As of September 16, 2025: https://www.darpa.mil/research/programs/atmospheric-water-extraction

[118] Atlantis Solar, "Atlantis H2O Elite Atmospheric Water Generator," product page, undated. As of September 16, 2025: https://www.atlantissolar.com/atlantis_h2o_elite.html

[119] Patrice Creel, U.S. Army Engineer Research and Development Center, "Water Recover Patent awarded to Construction Engineering Research Laboratory team," news release, 2022. As of September 16, 2025: https://www.erdc.usace.army.mil/Media/News-Stories/Article/2855465/water-recover-patent-awarded-to-construction-engineering-research-laboratory-te/

[120] Yaxuan Zhao, Weixin Guan, Yan Zhe Wong, and Guihua Yu, "Material-to-System Tailored Multilayer-Cyclic Strategy Toward Practical Atmospheric Water Harvesting," *Proceedings of the National Academy of Sciences*, Vol. 122, No. 20, May 12, 2025. As of September 16, 2025: https://doi.org/10.1073/pnas.2500928122

[121] Nathan Ortiz and Sameer Rao, "Review of Rain and Atmospheric Water Harvesting History and Technology," *Oxford Research Encyclopedia of Environmental Science*, March 20, 2024. As of September 16, 2025: https://doi.org/10.1093/acrefore/9780199389414.013.613

per liter compared with bulk water resupply. An example unit would be the Solaris AWG (also sold via Watergen A10 model) which produces around 10 L/day, retailing for approximately \$879–\$1,600 depending on the retailer and humidity level.[122] Larger systems for forward operating bases can cost tens of thousands of dollars but become cost-competitive in environments where fuel and bottled water resupply is logistically complex or high risk. Operational savings are likely when AWE offsets repeated convoy missions or airlift sorties, but those benefits depend on theater logistics and mission duration.

In contrast, combustion-based water recovery—using engine exhaust—has demonstrated water yields of 0.5–0.6 gallons per gallon of fuel burned, but such systems are large, maintenance-intensive, and platform-specific, making them less adaptable for ACE concepts. For the USAF, AWH's ability to produce potable water on-site could reduce reliance on vulnerable supply lines and enhance mission resilience in austere environments. However, further independent testing is needed to verify real-world power demands, reliability, and lifecycle costs before large-scale fielding.

Considerations for the USAF for the Employment of Atmospheric Water Harvesting Technology

AWH technology presents significant operational advantages for the USAF, particularly in austere and remote environments where access to traditional water infrastructure is limited. For USAF personnel operating in isolated or resource-scarce areas, AWH could serve as a critical lifeline, ensuring reliable access to potable water while reducing the need for resupply missions. However, rigorous testing for reliability and durability is essential to ensure these systems can withstand the demands of military operations. Continued investment in AWH technology aligns with the USAF's strategic priorities of enhancing operational resilience and reducing logistical complexity in future conflicts.

[122] Solaris WaterGen, "Solaris WaterGen A10 (USA / 110V) – 10 L/Day Atmospheric Water Generator," product page, 2024. As of September 16, 2025: https://www.solariswatergen.com/product/solaris-watergen-a10-for-usa-110-v/

Chapter 4. Conclusions

The technology survey presented in this report (and the previous survey from FY 2021) provide information on new and emerging technologies that could enhance support of ACE operations. Table 4.1 contains a summary of our updated technology survey. The columns highlight major findings from this analysis including advantages, drawbacks, and ROM costs for each of the technologies evaluated.

In general, lighter, leaner technologies like advanced photovoltaics, next-generation batteries, and atmospheric water harvesting devices that provide needed energy and water resources could enable operations at austere locations and provide resilience at established forward operating locations. Other technologies such as airborne wind turbines, fuel cells, and V2G systems could also be considered in the near term. Identifying promising technologies, however, is only one part of leveraging their capabilities for the DAF. There are acquisition and technology transition challenges in moving emerging technologies from innovation to fielding for warfighter use, discussed in more depth in Volume 1.

For example, AFWERX, within Air Force Materiel Command, has a well-defined way to investigate and invest in new and emerging technologies and when those new technologies are directly tied to weapon system acquisition, the process is clear—through the weapon system program office. However, when the new technology is not directly tied to a weapon system, the process for testing, acquiring, and fielding is less clear. Moreover, the U.S. Army serves as the executive agent for tactical generators and expeditionary equipment. For the USAF to leverage innovation and potential technologies tied to expeditionary basing from across services, the USAF needs to monitor Army Small Business Innovation Research (SBIR).

In Volume 1, we recommend the DAF establish a clear process for introducing new technologies into the USAF inventory, ensuring that innovation aligns with operational needs. This includes screening, assessing, validating, and fielding new technologies more efficiently. We also recommend that SAF/IEE monitor Army SBIR for innovations and potential technologies that could be leveraged by the USAF.

Table 4.1. Summary of Updated Technology Survey

Technology	Advantages	Drawbacks	Rough-Order-of-Magnitude Costs
Airborne wind turbines	• Reduce reliance on fuel resupply missions • Price competitive with similar-sized fixed wind turbines and lower than sustained fuel supply in remote theaters	• Airspace integration is a critical operational challenge • Potential for mechanical failures automatic descent mechanisms pose safety hazards • Upfront capital costs are high and require new training	• $1.7 million to $2.8 million as an overall investment
Photovoltaics	• Increases power supply portability and operation flexibility	• Without established maintenance or security provisions, systems fall at the risk of dysfunctionality • Permanent setups may require host-nation approval	• Initial investment of $400 to $1,000 per panel depending on manufacturer, power rating, and efficiency
Fuel cells	• Clean and scalable alternative method for energy production • Modular design supports rapid deployment and adaptability • High durability with relatively high efficiency	• Requires a secure and consistent fuel supply chain, which may not exist in remote or contested environments • Safety considerations with storage and transport because of volatility of fuel • Maintenance and technical support require skilled personnel and spare parts, adding operational footprint	• Typically ranges from $1,000 to $5000 for individual fuel cell. Full stack systems can range from $9,000 to $20,000. Liquid cooled fuel cell systems for high performance can cost anywhere between $80,000 to $200,000
Nuclear fusion	• Feasible implementation could significantly increase operational power reserves • Could significantly reduce logistical burdens while replacing fossil fuels as a primary energy source	• Cost for implementation is still relatively high • Reactor designs are complex which can lead to mismanagement or poor maintenance • Skilled personnel would be required for upkeep • Requires further development for feasible deployment	• Construction can range from $1 billion to $25+ billion per facility for an experimental fusion reactor
Portable nuclear reactors	• Reliable, carbon-free energy source that can operate continuously for years without refueling	• Costly • Host-nation approval is likely to be a significant obstacle • Physical security, layered defenses, and contingency planning for reactor	• Initial estimates for reactor can range from $50 million–$3 billion for small modular reactors, with an estimated levelized cost of energy of $5,000/kWh with current systems

Technology	Advantages	Drawbacks	Rough-Order-of-Magnitude Costs
	• Minimal maintenance would be required if fully trained personnel are present	shutdown or sabotage would need to be incorporated into ACE basing concepts	
Batteries	• High denser batteries can increase operational endurance • Reliability and performance under harsh conditions can enhance mission sustainment	• Lithium-based systems, even in safer chemistries, are hazardous for air transport—requires strict packaging and handling • Adoption requires new training, safety protocols, and end-of-life recycling processes	• For advanced lithium iron phosphate-based batteries, initial price ranges from $400–$900 per unit depending on the power density and configuration.
Ultracapacitors	• Ideal for applications involving transient loads, backup power, and cold-start operations • Low internal resistance and high reliability • Compact size, rapid recharge time, and broad operating temperature range	• Low energy density • High self-discharge rate • Relatively high cost per watt compared with batteries	• Depending on application, $100–$500 per small modules (low Farads, low voltage); medium modules (10–100 Farads, higher voltage) between $500–$2,000 per unit; large modules for custom deployment assemblies (>1000 Farads) between $2,000–$10,000+ per unit
Hydrogen energy storage	• Portable as a compressed gas, and has the highest energy per mass of any fuel • Can be refueled using locally generated renewable energy • Operate very quietly and produce little to no vibrations	• Current systems require large volume containers to store hydrogen • On-site production demands significant water and power resources • Transportation requires regulatory approval • Hydrogen storage systems are less mature and reliable for rugged military use • Capital and maintenance costs limit suitability for rapidly shifting operations	• $5,000–$1,000,000+ per system, with portable units at the lower end and containerized or base-scale systems at the higher end
Vehicle-to-grid systems	• Can provide instantaneous grid support such as load balancing, frequency control, congestion management, and voltage control which can leverage the use of EVs as a mobile power reserve	• Battery degradation may occur due to rapid charge/discharge cycles • High implementation costs for bidirectional chargers • Introduces higher data protection risks and requires additional layer of cyber protection • Possible grid overloading from multiple V2G systems can lead to instability	• Bi-directional vehicle-to-grid chargers are between $2,500–$12,000 each, with installation adding $600–$12,700 depending on site complexity. levelized cost of energy between $0.085–$0.243/kWh

Technology	Advantages	Drawbacks	Rough-Order-of-Magnitude Costs
Water filtration and decontamination	• Advanced filtration systems can increase the amount of potable water necessary for operations • Allows for the use of water sources local to the field of operations • Portable filtration units enhance mission sustainment and endurance	• Larger modular systems can be expensive • Would require regular maintenance for continued operations • Can be a health risk if chemical, biological, or radiological agents are not fully removed by standard filtration technologies	• Nano-filtration units capable of pricing 500–1,000 L/day range from $5,000–$20,000 depending on throughput, power integration and automation features. Larger systems for emergency deployment can range from $50,000 to over $200,000
Reverse osmosis	• Reduces hard water, minerals, and microbials that are harmful • Simple to operate so complex training is not necessary • Lower power demand than a comparable distillation plant	• Can generate a significant amount of water waste if single point of use system • Regular maintenance is required for continued effectiveness and operation • Would require a reliable power source for continued operation	• $1,000–$5,000 for unit producing 1,000 L/day and $15,000–$60,000 for units producing 5,000–20,000 L/day. Large, containerized RO plants, or base-level may exceed $100,000–$500,000
Capacitive deionization	• Operates at low pressure, consumes less energy, and can be regenerated for repeated cycles • Suitable for mid-range brackish water use • Can be used to complement RO and advanced filtration systems by reducing energy use	• Less effective for high salinity water • Can be degraded by the presence of organic contaminants, heavy metals, and particulates which are common in austere environments • Requires a stable electrical supply which may be challenging to maintain in dispersed or mobile operations • Best suited for small to medium scale water purification	• $13,000–$15,000 per unit producing between 10,000–50,000 L/day
Atmospheric water harvesting	• Reduces reliance on local rivers, lakes, or wells, which may be scarce, contaminated, or controlled by adversaries • Can provide potable water in remote or contested environments where no water sources may be present • Can be powered by solar energy sources which enhances sustainability and reduces dependence of fuel supplies	• Highly dependent on the humidity, which may not occur in dry operational theaters • Requires significant electrical power to condense and collect water, which may strain field energy resources or require additional power sources • Produces relatively small amounts of water, potentially insufficient for large groups or base-scale needs • Require regular maintenance, which may be challenging in rapid moving deployments • Harvested water may require additional purification steps due to airborne pollutions, dust, and chemical agents present in the air	• $1,500–$5,000 depending on unit producing between 5–20 L/day

Abbreviations

ACE	agile combat employment
AFCEC	Air Force Civil Engineering Center
AFRL	Air Force Research Lab
AFWERX	Air Force Work Project
Ah	Ampere-hours
AWE	atmospheric water extraction
AWH	atmospheric water harvesting
CUI	Controlled Unclassified Information
CDI	capacitive deionization
DAF	Department of Air Force
DAFC	direct ammonia fuel cell
DOW	Department of War
EPA	Environmental Protection Agency
EV	electric vehicle
FFRDC	federally funded research and development center
ITER	International Thermonuclear Experimental Reactor
k/m^3	kilogram per cubit meter
kWh	kilowatt-hour
LCOE	levelized cost of energy
LH_2	liquid hydrogen
MCDI	Membrane Capacitive Deionization
MWh	megawatt-hour
NRC	Nuclear Regulatory Commission
NREL	National Renewable Energy Laboratory
O&M	operation and maintenance
PACAF	Pacific Air Forces
PAF	Project AIR FORCE
PV	photovoltaics
RO	reverse osmosis
ROM	rough order of magnitude
SAF/IEE	Secretary of Air Force – Installations, Energy, and the Environment
SBIR	Small Business Innovation Research
SDI	Smart Deionization
SMR	small modular reactors

SOFC	solid oxide fuel cells
UAV	unmanned aerial vehicle
USAF	U.S. Air Force
UV	ultraviolet
V2G	vehicle to grid

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